METHOD 01

Pearson Correlation

Question: How strong is the linear link between two return series?

What this method is

Pearson r measures the linear co-movement of two series. A value near +1 means the returns rise and fall together on a straight line. A value near -1 means they move in opposite directions.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: one full-sample Pearson r, two-sided p, and a Fisher 95% band. Output unit: correlation in [-1, 1].

What you get: You get the average linear link over the full history for each unordered pair.
What you do not get: You do not get crash risk, lead-lag, or a trade signal.

Formula

```
r = SUM((x - x_bar)(y - y_bar))
/ sqrt( SUM((x - x_bar)^2) * SUM((y - y_bar)^2) )
```

Read it: Put your two return series into the formula. x, y = Daily log returns of the two FX pairs; x_bar, y_bar = Sample means of those returns; r = Full-sample linear correlation. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
x, y	Daily log returns of the two FX pairs
x_bar, y_bar	Sample means of those returns
r	Full-sample linear correlation

Pluses

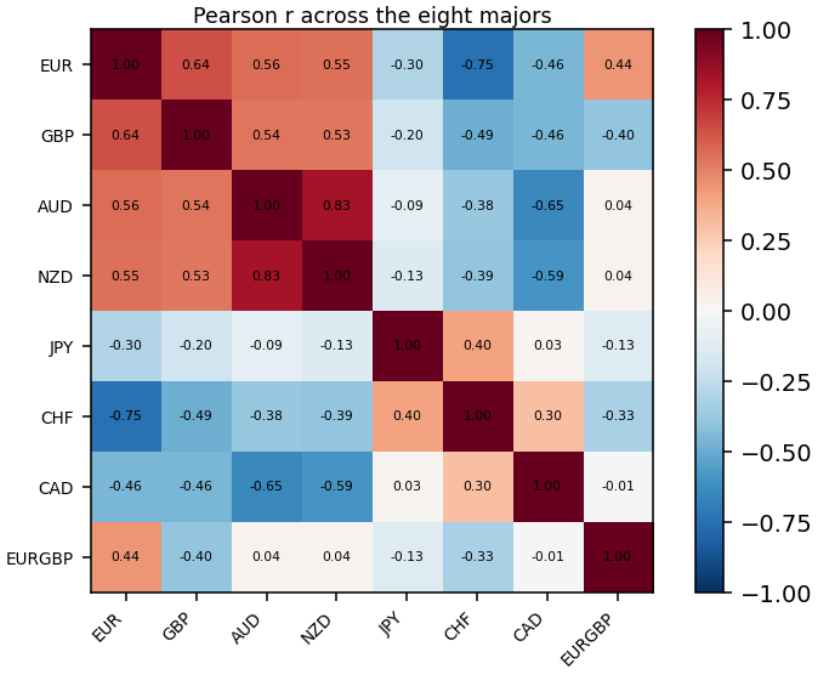
- The formula is simple and fast.
- Many desks use this as the base measure.
- You can report r with a p-value and a band.

Minuses

- It finds only linear links.
- Outliers can move r a large amount.
- One full-sample r hides regime changes.

Use it for: A first screen of pair co-movement. A clear baseline for other methods.
Do not use it for: Crash risk or non-linear links. Lead-lag or who drives whom.

Chart for this method



Portfolio heatmap of full-sample Pearson r. Strongest |r|: AUDUSD-NZDUSD = 0.829.

METHOD 02

Spearman Rank Correlation

Question: Do the ranks of the two series move together?

What this method is

Spearman correlation applies Pearson r to the ranks of the returns. It tracks monotone links. It is less sensitive to a few extreme days.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: Spearman rho on ranks, with Pearson r as a side check. Output unit: rank correlation in [-1, 1].

What you get: You get a robust monotone co-movement score for each pair.
What you do not get: You do not get time-varying or directional dependence.

Formula

```
rho_s = Corr( rank(x), rank(y) )
```

Read it: Put your two return series into the formula. rank(x) = Order of each return in the sample; rho_s = Pearson correlation of the two rank series. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
rank(x)	Order of each return in the sample
rho_s	Pearson correlation of the two rank series

Pluses

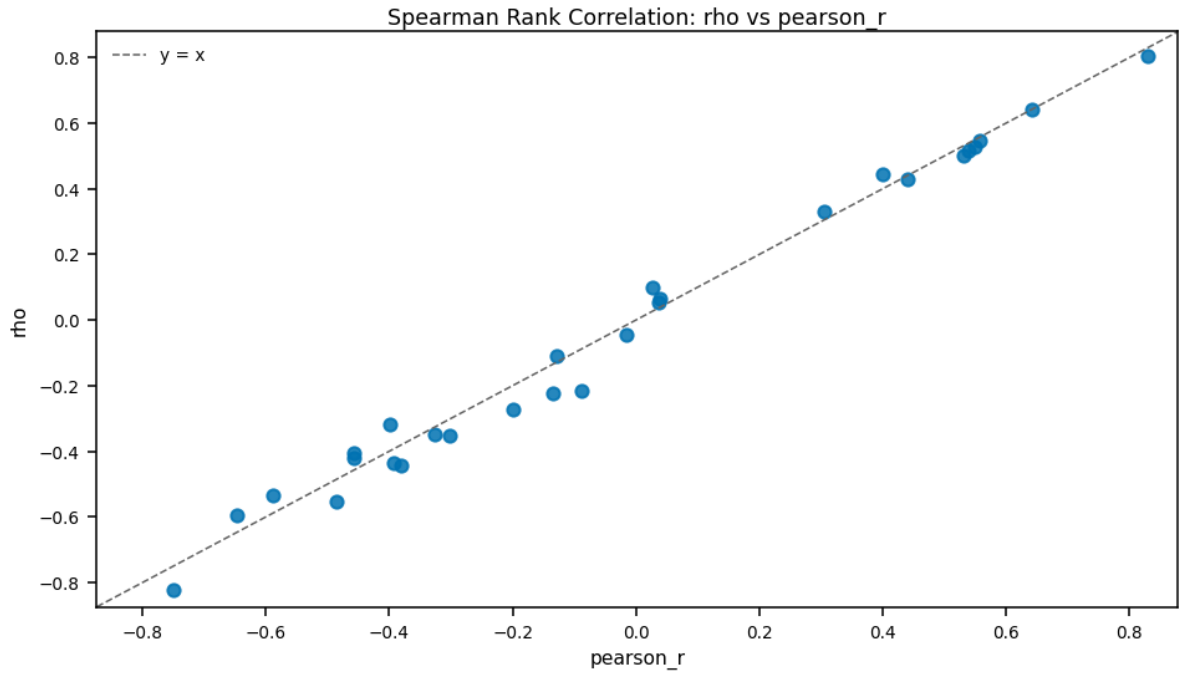
- Ranks reduce the effect of extreme returns.
- It finds monotone links, not only linear ones.
- It stays close to Pearson when the link is linear.

Minuses

- It still averages the full sample.
- Ties need a clear rank rule.
- It does not show time variation.

Use it for: A robust check next to Pearson r. Cases with heavy tails in returns.
Do not use it for: Dynamic regime monitoring alone. Directional lead-lag claims.

Chart for this method



Each point is one FX pair. Largest |rho – pearson_r|: AUDUSD-USDJPY (-0.214 vs -0.088).

METHOD 03

Kendall Tau-b

Question: How often do pair-of-days move in the same order?

What this method is

Kendall tau-b counts concordant and discordant day pairs. It measures rank agreement with a tie correction.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: tau-b with p-value; Pearson r as a side check. Output unit: tau-b in [-1, 1].

What you get: You get a concordance-based rank link for each pair.
What you do not get: You do not get a dynamic path or a crash frequency.

Formula

```
tau_b = (C - D) / sqrt( (C + D + T_x) * (C + D + T_y) )
```

Read it: Put your two return series into the formula. C = Concordant day-pairs; D = Discordant day-pairs; T_x, T_y = Tie corrections on each leg. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
C	Concordant day-pairs
D	Discordant day-pairs
T_x, T_y	Tie corrections on each leg

Pluses

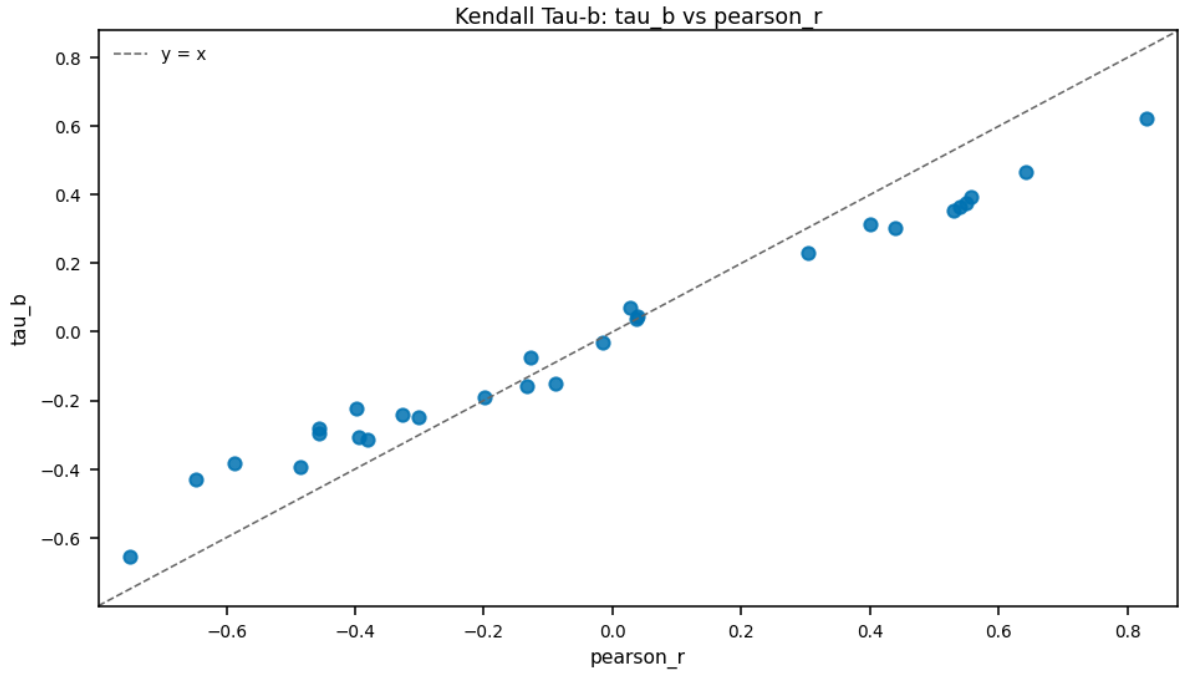
- It uses concordant and discordant pairs.
- Tau-b handles ties in a standard way.
- It is robust to some outliers.

Minuses

- The count grows with n squared cost.
- The scale differs from Pearson.
- It is still a static full-sample score.

Use it for: Rank agreement checks beside Spearman. Samples where ties appear.
Do not use it for: Intraday microstructure alone. Crash co-occurrence counts.

Chart for this method



Each point is one FX pair. Largest |tau_b – pearson_r|: AUDUSD-USDCAD (-0.430 vs -0.647).

METHOD 04

Robust Clipped Correlation

Question: What is Pearson r after you clip large returns?

What this method is

Clipped correlation caps each return at +/- c, then computes Pearson r. The cap limits the pull of jump days.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: clipped rho at c in {{0.01, 0.02, 0.03}}; headline uses c=0.02. Output unit: correlation in [-1, 1].

What you get: You get a Pearson-style score that is less driven by a few jumps.
What you do not get: You do not get a measure of the extremes themselves.

Formula

```
x_tilde = min(c, max(-c, x))
y_tilde = min(c, max(-c, y))
rho_clip = Corr(x_tilde, y_tilde)
```

Read it: Put your two return series into the formula. c = Clip level (0.02 in the headline); x_tilde, y_tilde = Returns after the cap; rho_clip = Pearson r on clipped returns. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
c	Clip level (0.02 in the headline)
x_tilde, y_tilde	Returns after the cap
rho_clip	Pearson r on clipped returns

Pluses

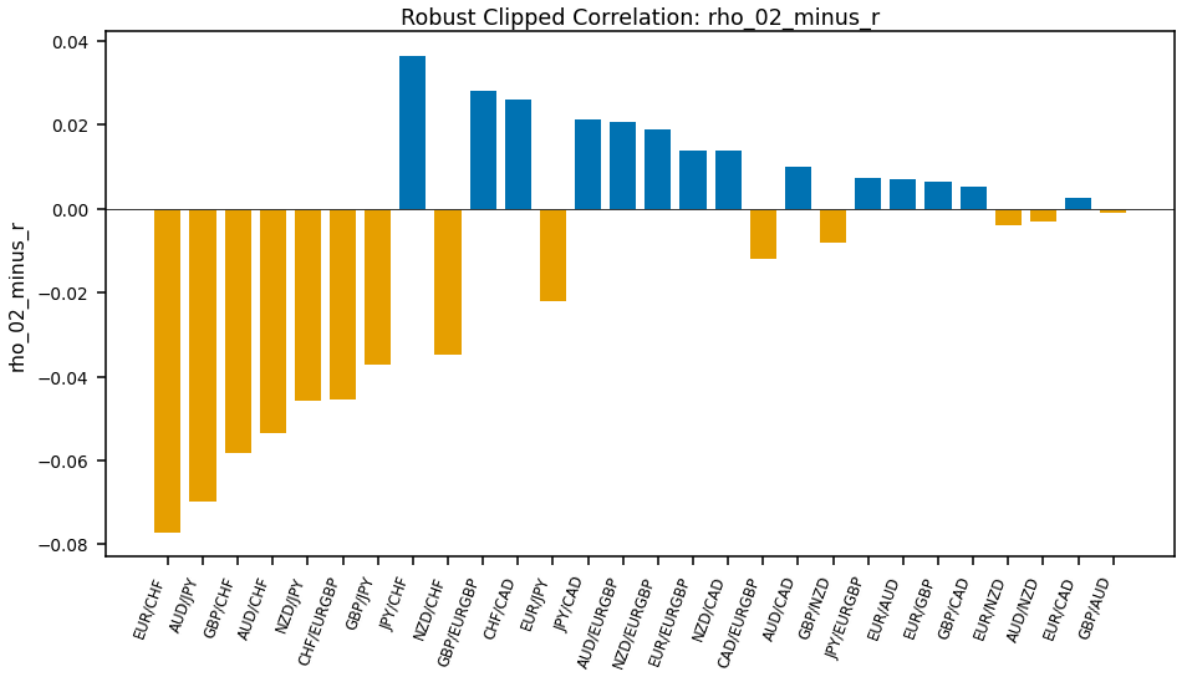
- Clipping limits the effect of jump days.
- You keep a Pearson-style scale.
- You can count how often the cap hits.

Minuses

- The cap c is a free parameter.
- Clipping can hide true crash links.
- It remains a static average.

Use it for: Sanity check when one day dominates r. Compare rho_clip with raw Pearson r.
Do not use it for: Tail dependence measurement. Cases where extremes are the signal.

Chart for this method



Gap chart (method-specific). Largest |rho_02_minus_r|: EURUSD-USDCHF = -0.077.

METHOD 05

Rolling Correlation

Question: How does Pearson r change in windows of 20, 60 and 100 days?

What this method is

Rolling correlation recomputes Pearson r on the last W days. It shows how the linear link changes through time.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: latest r for W in {{20, 60, 100}} on a 17-pair watchlist, plus a full-sample baseline from method 01. Output unit: correlation in [-1, 1].

What you get: You get short-horizon r paths and gaps versus the long baseline.
What you do not get: You do not get a volatility-adjusted or asymmetric path.

Formula

```
rho_t^(W) = Corr( last W observations of x and y )
```

Read it: Put your two return series into the formula. W = Window length in trading days; rho_t^(W) = Pearson r on the window that ends at t. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
W	Window length in trading days
rho_t^(W)	Pearson r on the window that ends at t

Pluses

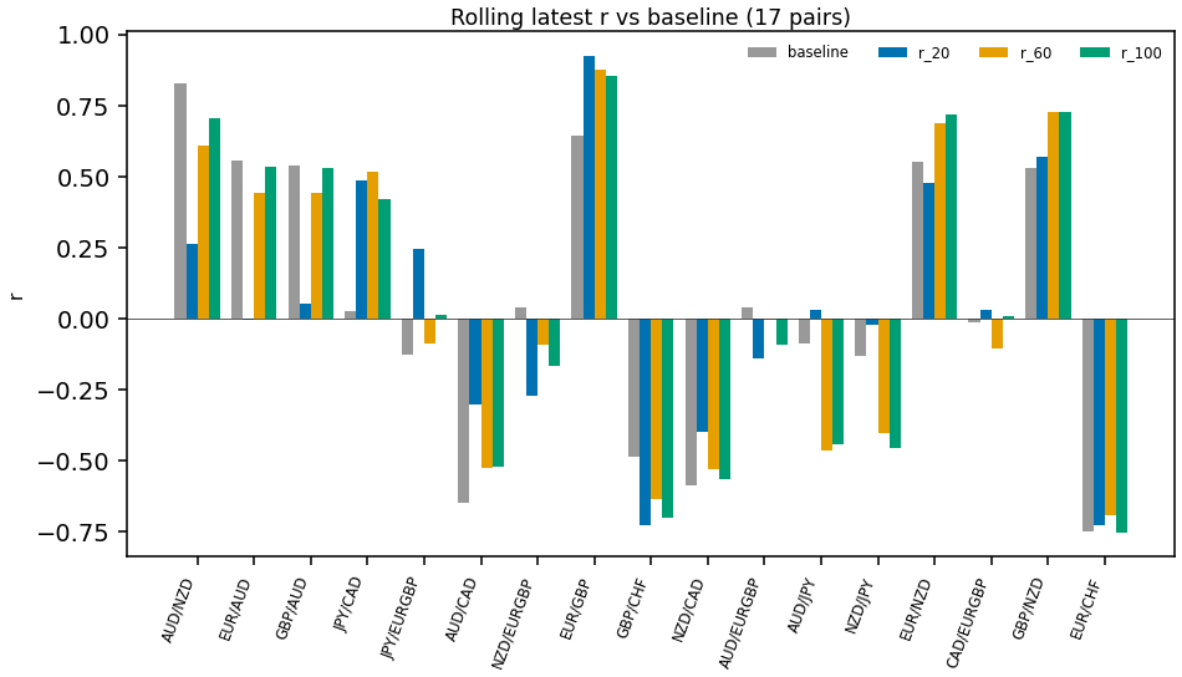
- It shows short-horizon breaks vs the baseline.
- Windows 20 / 60 / 100 are easy to read.
- Desks already watch rolling r.

Minuses

- Equal weights create a ghost drop at the window edge.
- High volatility can inflate rolling r.
- Short windows are noisy.

Use it for: Watchlist alerts when r_20 moves far from baseline. A simple time-varying screen.
Do not use it for: Volatility-adjusted dependence alone. Formal early-warning claims without tests.

Chart for this method



Rolling windows vs baseline. Largest |r_20 – baseline|: AUDUSD-NZDUSD (r_20=0.263, baseline=0.829).

METHOD 06

EWMA Correlation

Question: What is the latest exponentially weighted correlation?

What this method is

EWMA correlation gives more weight to recent days. It updates every day without a hard window edge.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: latest EWMA rho path value; Pearson r as a side check. Output unit: correlation in [-1, 1].

What you get: You get a smooth, recent-weighted linear dependence score.
What you do not get: You do not get GARCH residual standardisation or crash asymmetry.

Formula

```
c_t = lam*c_{t-1} + (1-lam)*x_t*y_t
v_x,t = lam*v_x,t-1 + (1-lam)*x_t^2
rho_t = c_t / sqrt(v_x,t * v_y,t)
```

Read it: Put your two return series into the formula. lam = Decay factor (near 0.94 in RiskMetrics style); c_t = EWMA cross-product; rho_t = EWMA correlation at day t. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
lam	Decay factor (near 0.94 in RiskMetrics style)
c_t	EWMA cross-product
rho_t	EWMA correlation at day t

Pluses

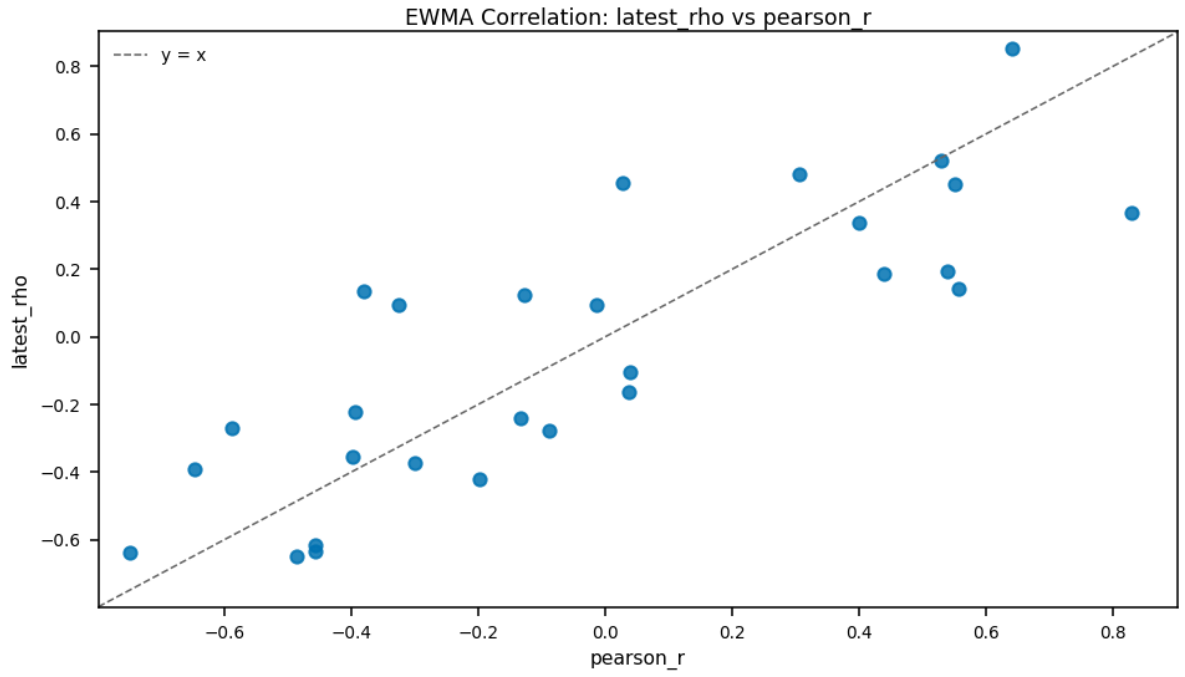
- Recent days get more weight than old days.
- No hard window edge like rolling W.
- It updates every day with one pass.

Minuses

- Lambda is a free parameter.
- It does not standardise GARCH residuals.
- It treats up and down days the same.

Use it for: A smooth alternative to rolling Pearson. Daily monitoring of latest_rho vs Pearson r.
Do not use it for: Crash asymmetry without ADCC. Long-run structural levels alone.

Chart for this method



Each point is one FX pair. Largest |latest_rho – pearson_r|: AUDUSD-USDCHF (0.135 vs -0.381).

METHOD 07

Partial and Residual Correlation

Question: What link remains after you remove a USD strength factor?

What this method is

Partial correlation removes a common control factor, then correlates the residuals. In this project the control is a leave-currency-out USD strength factor.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: OLS residual Pearson rho after control z; also R-squared of each leg on z. Output unit: residual correlation in [-1, 1].

What you get: You get the cross-currency link that remains after the dollar factor.
What you do not get: You do not get a dynamic or tail-risk measure.

Formula

```
x = a + b*z + e_x
y = c + d*z + e_y
rho_partial = Corr(e_x, e_y)
```

Read it: Put your two return series into the formula. z = Leave-currency-out USD strength factor; e_x, e_y = OLS residuals after removing z; rho_partial = Pearson r of the residuals. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
z	Leave-currency-out USD strength factor
e_x, e_y	OLS residuals after removing z
rho_partial	Pearson r of the residuals

Pluses

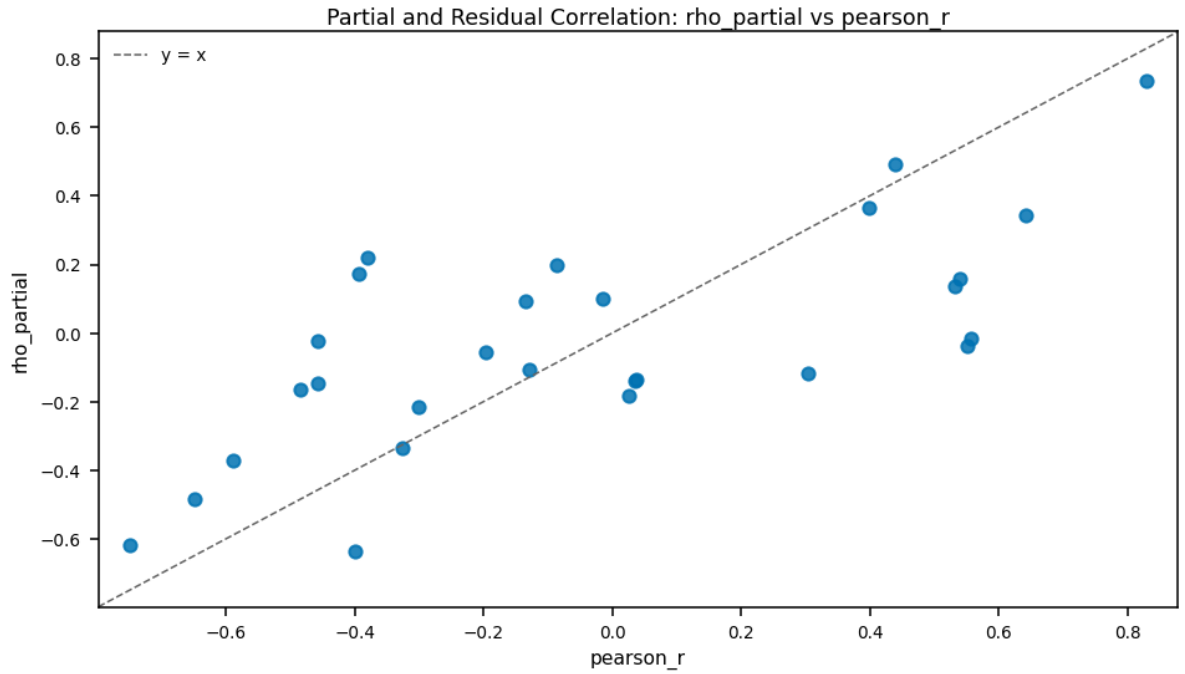
- It strips a shared dollar factor from majors.
- Large gaps show fake USD-driven Pearson links.
- R-squared on z shows how much USD explains.

Minuses

- The control z is a model choice.
- Bad controls can create false residuals.
- It is still a static residual Pearson.

Use it for: Cross-currency links beyond the dollar. Checks that EUR and AUD are not only USD moves.
Do not use it for: Pairs with no clear common factor. Intraday microstructure alone.

Chart for this method



Each point is one FX pair. Largest |rho_partial – pearson_r|: AUDUSD-USDCHF (0.219 vs -0.380).

METHOD 08

Shrinkage Correlation

Question: What correlation matrix do you get after Ledoit-Wolf shrinkage?

What this method is

Shrinkage blends the sample covariance toward a simple target. It reduces noise in a correlation matrix used for portfolios or PCA.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: Ledoit-Wolf shrunk pair correlation rho_LW and intensity delta_LW. Output unit: correlation in [-1, 1].

What you get: You get a quieter pair correlation for matrix-based work.
What you do not get: You do not get dynamics, tails, or lead-lag.

Formula

```
Sigma_tilde = (1 - delta)*S + delta*mu*I
rho_ij = Sigma_tilde_ij / sqrt(Sigma_ii * Sigma_jj)
```

Read it: Put your two return series into the formula. S = Sample covariance; delta = Shrinkage intensity (Ledoit-Wolf); rho_ij = Correlation from the shrunk matrix. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
S	Sample covariance
delta	Shrinkage intensity (Ledoit-Wolf)
rho_ij	Correlation from the shrunk matrix

Pluses

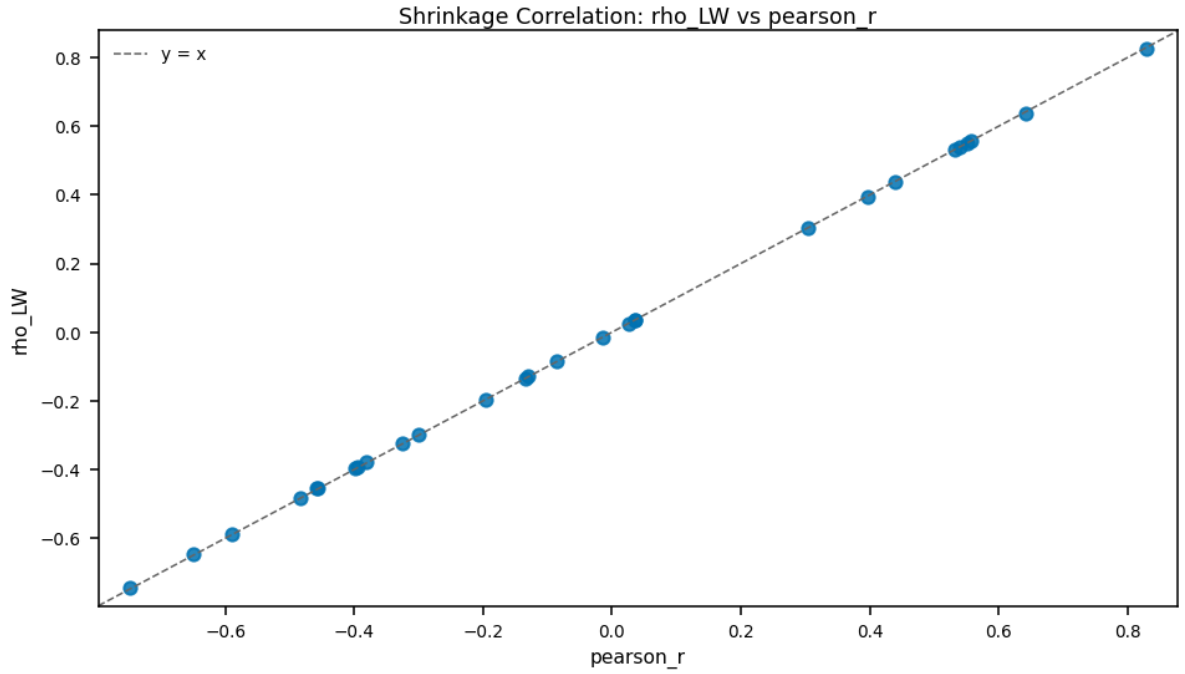
- Shrinkage reduces noise in large matrices.
- Delta has a data-driven Ledoit-Wolf rule.
- It helps portfolio and PCA inputs.

Minuses

- Pair scores move toward a target.
- It can hide true large correlations.
- It does not add dynamics.

Use it for: Stable inputs to network or PCA work. Noise reduction when n is limited vs assets.
Do not use it for: Tail event frequency. Lead-lag prediction alone.

Chart for this method



Each point is one FX pair. Largest |rho_LW – pearson_r|: EURUSD-USDCHF (-0.744 vs -0.747).

METHOD 09

DCC-GARCH

Question: How does correlation evolve after GARCH standardisation?

What this method is

DCC-GARCH first filters each return with GARCH volatility. It then builds a dynamic correlation path on the standardised residuals.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: latest DCC rho after constant-mean GARCH(1,1) z-scores. Output unit: dynamic correlation in [-1, 1].

What you get: You get a vol-filtered, time-varying linear correlation.
What you do not get: You do not get downside asymmetry (that needs ADCC).

Formula

```
z_i,t = eps_i,t / sigma_i,t
q_ij,t = (1-a-b)*q_bar + a*z_i,t-1*z_j,t-1 + b*q_ij,t-1
rho_ij,t = q_ij,t / sqrt(q_ii,t * q_jj,t)
```

Read it: Put your two return series into the formula. z_i,t = GARCH-standardised residual; a, b = News and persistence weights; rho_ij,t = Dynamic conditional correlation. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
z_i,t	GARCH-standardised residual
a, b	News and persistence weights
rho_ij,t	Dynamic conditional correlation

Pluses

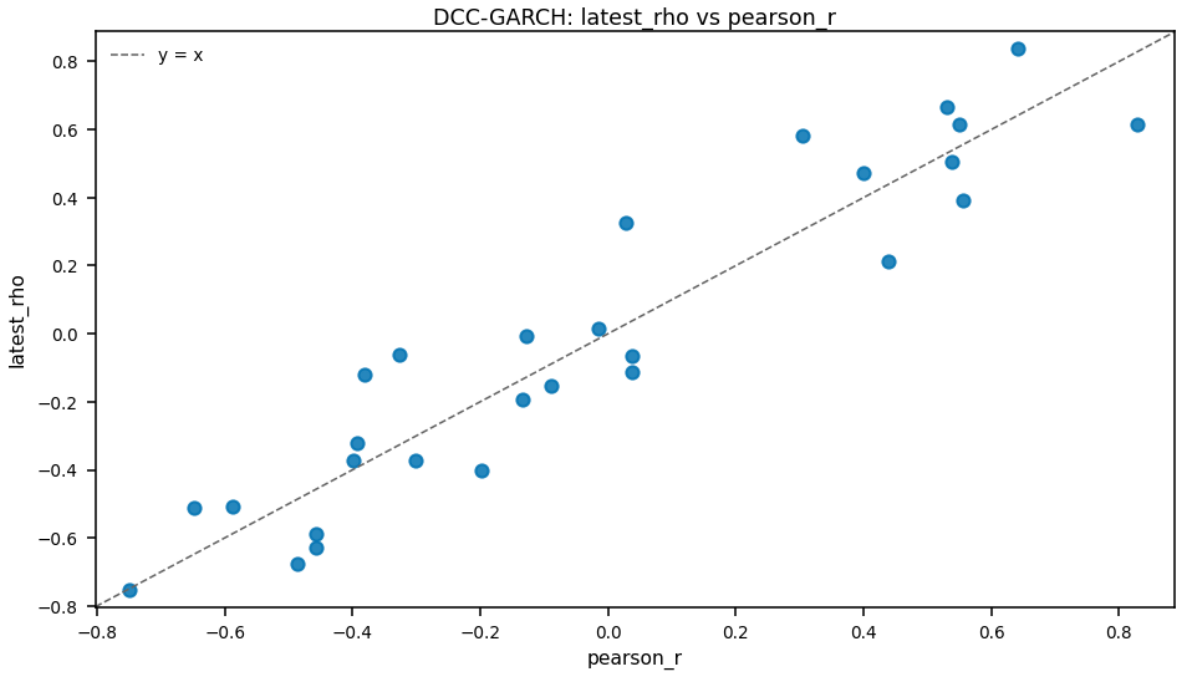
- It removes volatility clustering before correlation.
- Correlation updates every day.
- a and b give persistence of the path.

Minuses

- Fits can fail or hit bounds.
- Symmetric DCC treats crashes and rallies alike.
- Needs stable GARCH margins.

Use it for: Dynamic correlation after vol filtering. Comparison of latest_rho with Pearson r.
Do not use it for: Asymmetry in down markets (use ADCC). Tiny samples where GARCH fails.

Chart for this method



Each point is one FX pair. Largest |latest_rho – pearson_r|: USDJPY-USDCAD (0.325 vs 0.028).

METHOD 10

Asymmetric DCC

Question: Does correlation rise more after joint negative shocks?

What this method is

Asymmetric DCC extends DCC with a crash term g. Joint negative shocks can raise correlation more than joint positive shocks.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: latest ADCC rho, nested DCC, and parameters a, b, g. Output unit: dynamic correlation in [-1, 1]; g is an asymmetry weight.

What you get: You get a dynamic path that can react more to joint sell-offs.
What you do not get: You do not get a valid path when the optimiser fails to converge.

Formula

```
n_i,t = min(z_i,t, 0)
q_ij,t = (1-a-b-g)*Qbar + a*z_i*z_j + g*n_i*n_j + b*q_ij
rho_ij,t = q_ij,t / sqrt(q_ii,t * q_jj,t)
```

Read it: Put your two return series into the formula. n_i,t = Negative part of the residual (crash side); g = Asymmetry weight on joint downs; rho_ij,t = Asymmetric DCC correlation. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
n_i,t	Negative part of the residual (crash side)
g	Asymmetry weight on joint downs
rho_ij,t	Asymmetric DCC correlation

Pluses

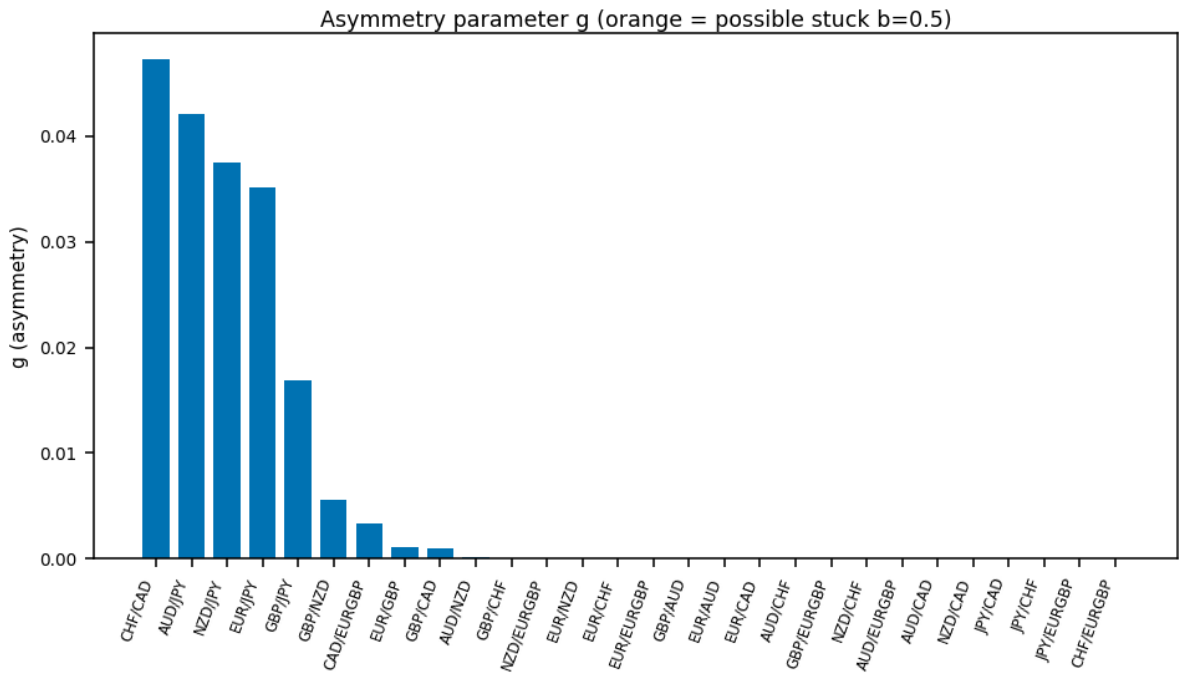
- The g term captures crash asymmetry.
- It nests symmetric DCC when g = 0.
- Useful for carry and safe-haven pairs.

Minuses

- More parameters than DCC.
- Optimizer bounds can stick (g=0 or bad b).
- Needs a likelihood-ratio check for g.

Use it for: Stress asymmetry vs calm periods. Gaps ADCC minus Pearson and ADCC minus DCC.
Do not use it for: Claims without fit diagnostics. Price-level mean reversion.

Chart for this method



ADCC asymmetry g. Largest g: USDCHF-USDCAD = 0.0473. Stuck b=0.5 fit: EURUSD-USDCHF.

METHOD 11

Regime-Conditioned Correlation

Question: How does correlation differ in quiet vs stress states?

What this method is

This method splits days into quiet and stress using a shared turbulence rule. It then computes Pearson r in each state.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: rho_quiet, rho_stress, and stress_minus_quiet from a look-ahead-free FX turbulence rule. Output unit: correlation in [-1, 1]; gap is a difference of correlations.

What you get: You get how linear co-movement changes when the market is stressed.
What you do not get: You do not get a volatility-bias-free stress rho without further adjustment.

Formula

```
rho_quiet = Corr(x,y | state=quiet)
rho_stress = Corr(x,y | state=stress)
```

Read it: Put your two return series into the formula. state = Quiet or stress from past turbulence RV; rho_quiet = Pearson r on quiet days; rho_stress = Pearson r on stress days. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
state	Quiet or stress from past turbulence RV
rho_quiet	Pearson r on quiet days
rho_stress	Pearson r on stress days

Pluses

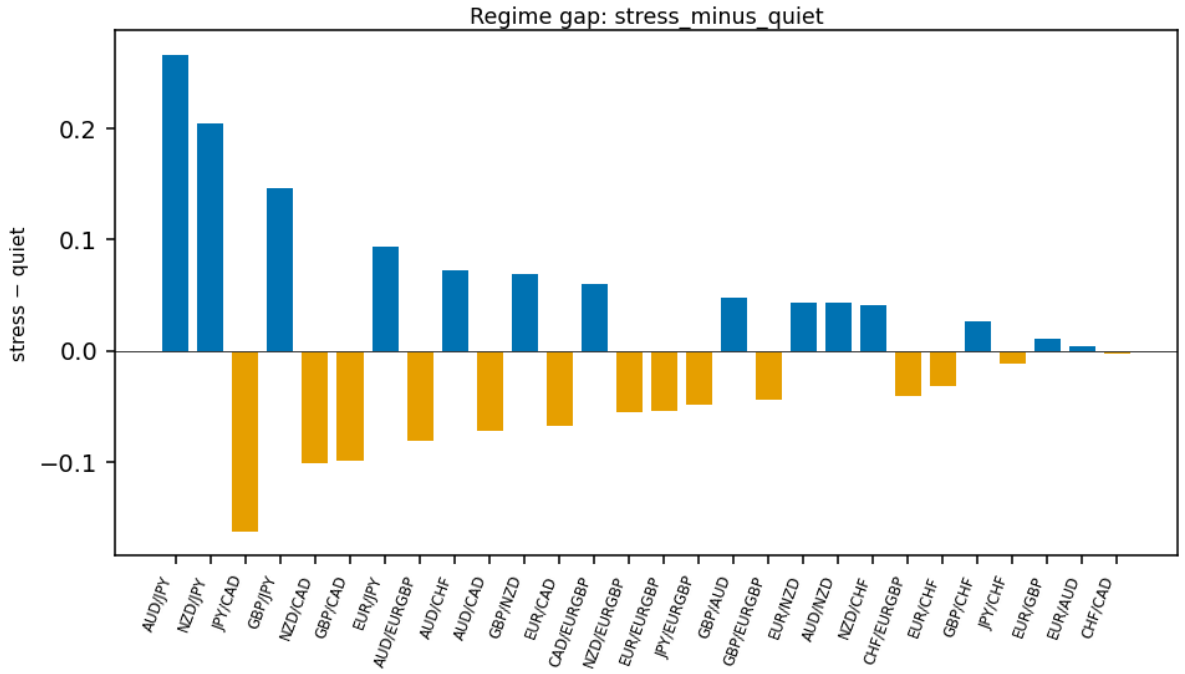
- It splits the sample by a turbulence rule.
- stress_minus_quiet flags regime flips.
- The state rule uses past data only.

Minuses

- High vol can bias raw stress correlation.
- State labels depend on the turbulence rule.
- Fewer stress days reduce power.

Use it for: Quiet vs stress co-movement maps. Carry pairs that flip in risk-off.
Do not use it for: Unadjusted stress gaps as final truth. Exact crash-date prediction.

Chart for this method



Stress minus quiet correlation. Largest |gap|: AUDUSD-USDJPY = 0.267.

METHOD 12

Lagged Cross-Correlation

Question: Does one return series lead the other at lag k?

What this method is

Lagged cross-correlation is Pearson r between one series today and the other series k days later. It is a simple lead-lag screen.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: rho at lag k=1 in both directions; also same-day rho_k0. Output unit: correlation in [-1, 1].

What you get: You get a signed lag-1 linear link in each direction.
What you do not get: You do not get proof of economic causality.

Formula

```
rho_{X->Y}(k) = Corr( x_t, y_{t+k} )
```

Read it: Put your two return series into the formula. k = Lag in trading days (headline uses 1); rho_{X->Y}(k) = Correlation of x today with y after k days. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
k	Lag in trading days (headline uses 1)
rho_{X->Y}(k)	Correlation of x today with y after k days

Pluses

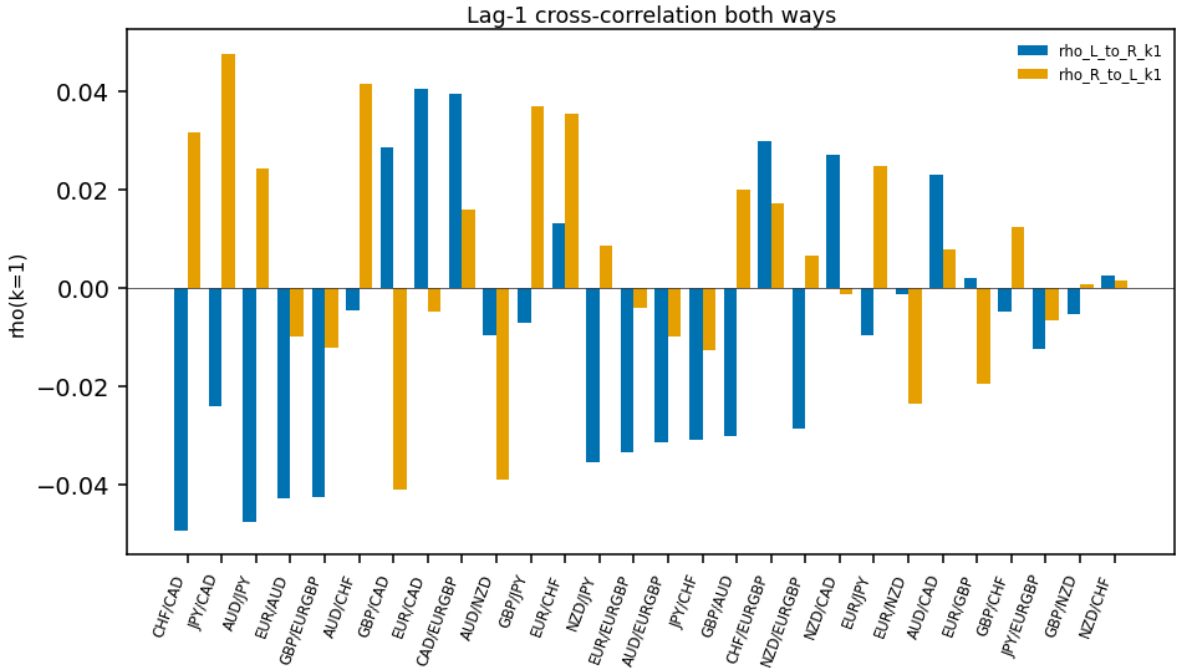
- It is a simple lead-lag screen.
- You can scan several lags k.
- k=0 recovers same-day Pearson.

Minuses

- Correlation is not causation.
- Many lags raise false positives.
- FX majors often show weak lags on D1.

Use it for: First look at directed timing. Compare with Granger and transfer entropy.
Do not use it for: Trading without out-of-sample tests. Claims of economic lead without costs.

Chart for this method



Directed scores. Peak |value|: USDCHF-USDCAD (rho_L_to_R_k1=-0.0494, rho_R_to_L_k1=0.0317).

METHOD 13

Granger Predictability

Question: Does past X help forecast Y beyond past Y?

What this method is

Granger predictability tests whether past X improves a forecast of Y after you already use past Y. A small p-value rejects the null of no incremental content.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: lag-1 and lag-5 F-test p-values in both directions. Output unit: p-value in (0, 1]; smaller means stronger rejection.

What you get: You get a directed predictability test for each ordered pair.
What you do not get: You do not get true structural causality or a trade rule.

Formula

```
y_t = c + a*y_{t-1} + b*x_{t-1} + e_t
H0: b = 0
```

Read it: Put your two return series into the formula. b = Coefficient on lagged X; H0 = Null: past X adds nothing; p = p-value of the test. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
b	Coefficient on lagged X
H0	Null: past X adds nothing
p	p-value of the test

Pluses

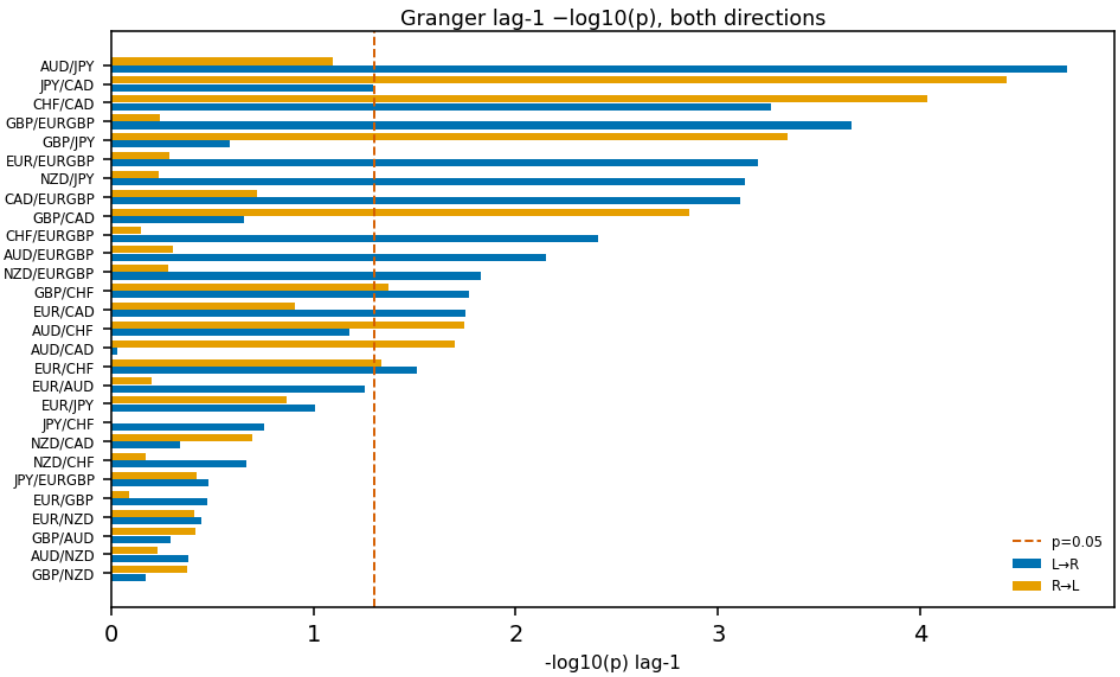
- It tests incremental forecast content.
- F-tests and p-values are standard.
- You can raise lag order p.

Minuses

- Reject H0 is not true causality.
- Multiple pairs need multiplicity control.
- Linear VAR misses non-linear leads.

Use it for: Directed predictability screens. Comparison with lagged correlation.
Do not use it for: Sole proof of economic causality. Non-linear information flow alone.

Chart for this method



Horizontal bars of −log10(p). Smallest lag-1 L→R p: AUDUSD-USDJPY = 1.88e-05.

METHOD 14

Distance Correlation

Question: Is there any dependence, including non-linear links?

What this method is

Distance correlation measures dependence from pairwise distances. In the population, dCor = 0 only under independence.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: biased sample dCor; also dCor minus |Pearson r|. Output unit: dCor in [0, 1].

What you get: You get a general dependence score and a non-linear excess over |r|.
What you do not get: You do not get a signed hedge ratio or a lead.

Formula

```
dCor = sqrt( mean(A~ * B~)
/ sqrt( mean(A~^2) * mean(B~^2) ) )
```

Read it: Put your two return series into the formula. A~, B~ = Double-centered distance matrices; dCor = Distance correlation in [0, 1]. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
A~, B~	Double-centered distance matrices
dCor	Distance correlation in [0, 1]

Pluses

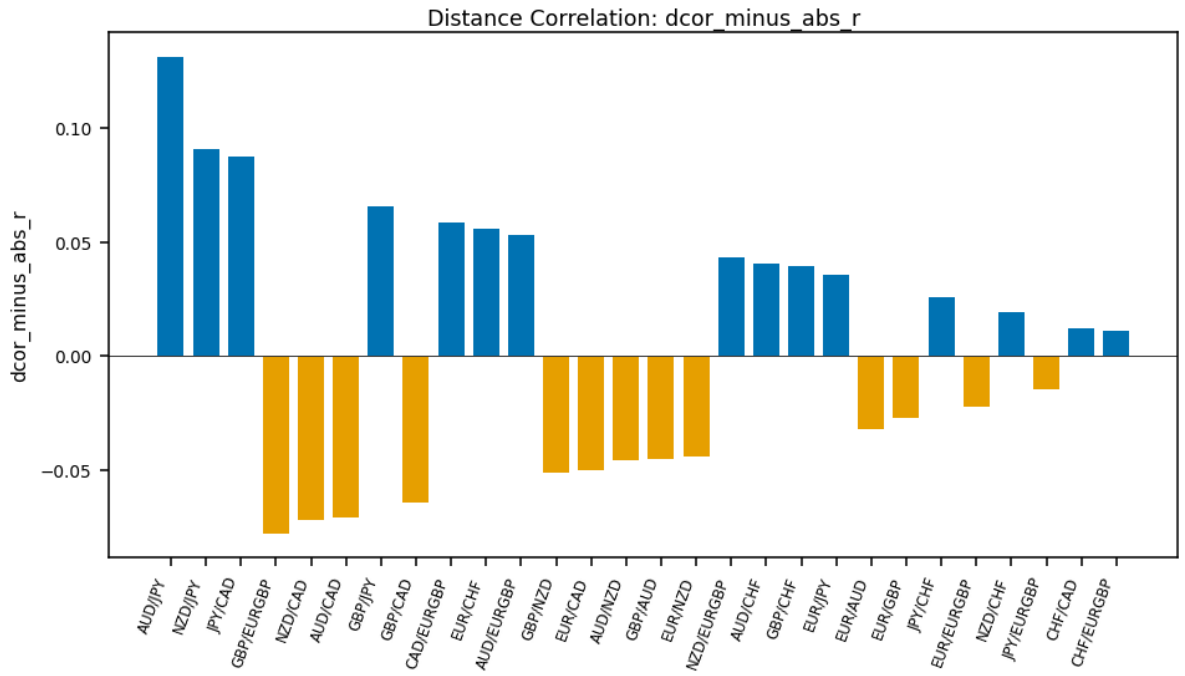
- dCor = 0 only under independence.
- It finds non-linear co-movement.
- Gap dCor - |r| flags extra dependence.

Minuses

- Computation uses distance matrices.
- Biased sample form needs care for tests.
- It does not give a signed direction.

Use it for: Pairs where Pearson r is near zero. Non-linear spillover checks.
Do not use it for: Signed hedge ratios. Lead-lag without extra tools.

Chart for this method



Gap chart (method-specific). Largest |dcor_minus_abs_r|: AUDUSD-USDJPY = 0.131.

METHOD 15

Mutual Information

Question: How much shared information do the two series hold?

What this method is

Mutual information measures shared information between binned returns. Normalised MI (NMI) puts the score on a comparable scale.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: discrete MI and normalised MI from equal-frequency bins. Output unit: NMI typically in [0, 1].

What you get: You get an information-theoretic dependence score for each pair.
What you do not get: You do not get a signed correlation for hedge sizing.

Formula

```
I(X;Y) = SUM p_ab ln( p_ab / (p_a * p_b) )
NMI = I / sqrt( H(X) H(Y) )
```

Read it: Put your two return series into the formula. p_ab = Joint bin probability; I(X;Y) = Mutual information in nats; NMI = Normalised mutual information. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
p_ab	Joint bin probability
I(X;Y)	Mutual information in nats
NMI	Normalised mutual information

Pluses

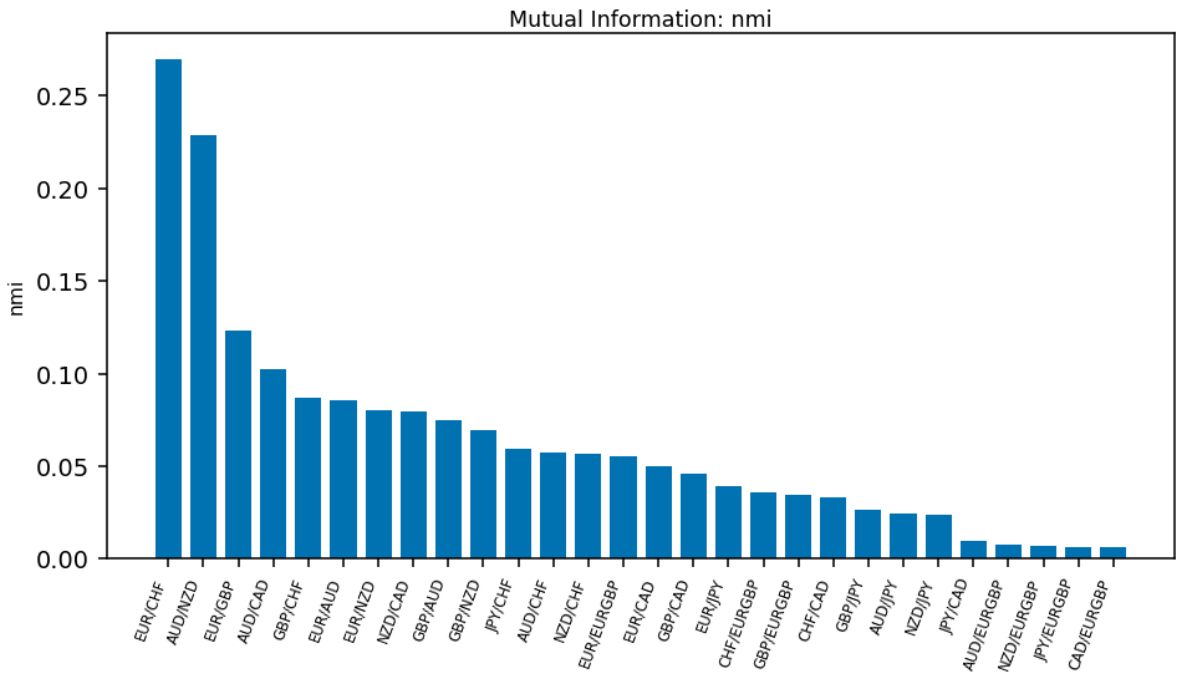
- It captures general dependence.
- NMI is easier to compare across pairs.
- It does not assume a linear form.

Minuses

- Bin count changes the estimate.
- Plug-in MI has positive bias.
- Scale differs from correlation.

Use it for: Information-theoretic dependence screens. Complement to distance correlation.
Do not use it for: Direct hedge sizing like Pearson r. Fine claims without bias correction.

Chart for this method



Sorted nmi. Highest: EURUSD-USDCHF = 0.270. Lowest: USDCAD-EURGBP = 0.006.

METHOD 16

HSIC Kernel Dependence

Question: Do kernel embeddings of X and Y show dependence?

What this method is

HSIC tests dependence in a reproducing kernel Hilbert space. A normalised HSIC helps rank pairs.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: HSIC and normalised HSIC with RBF bandwidths ell_x, ell_y. Output unit: normalised HSIC (dimensionless, larger means more dependence).

What you get: You get a flexible kernel dependence score.
What you do not get: You do not get an easy signed hedge ratio.

Formula

```
K_ij = exp( -(x_i-x_j)^2 / (2 ell^2) )
HSIC = tr(K H L H) / n^2
```

Read it: Put your two return series into the formula. ell = RBF bandwidth; K, L = Kernel matrices for X and Y; HSIC = Hilbert-Schmidt independence criterion. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
ell	RBF bandwidth
K, L	Kernel matrices for X and Y
HSIC	Hilbert-Schmidt independence criterion

Pluses

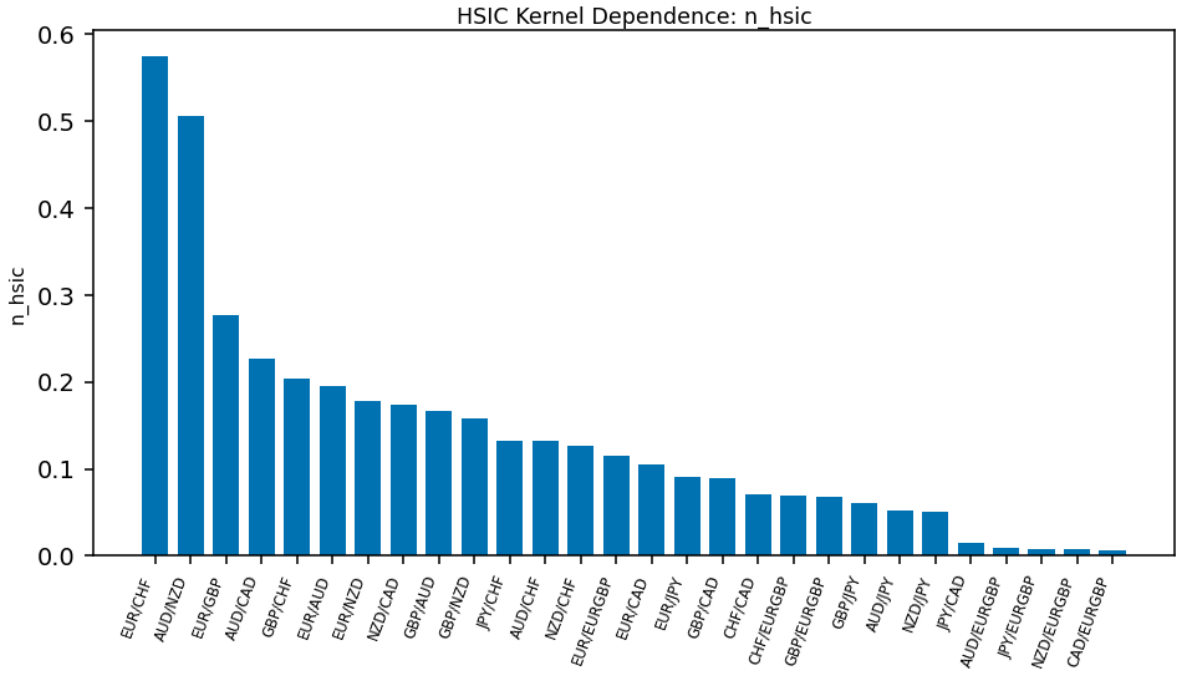
- Kernels detect flexible non-linear links.
- HSIC = 0 under independence (population).
- Normalised HSIC helps cross-pair rank.

Minuses

- Bandwidth ell is a free choice.
- Cost grows with n squared.
- Harder to explain than Pearson r.

Use it for: Kernel dependence next to dCor and MI. Cases with complex scatter shapes.
Do not use it for: Simple signed correlation reports. Very large n without approx methods.

Chart for this method



Sorted n_hsic. Highest: EURUSD-USDCHF = 0.575. Lowest: USDCAD-EURGBP = 0.007.

METHOD 17

Transfer Entropy

Question: Does past X add information about next Y beyond past Y?

What this method is

Transfer entropy measures directed information flow. It asks whether past X helps predict the next state of Y after you use past Y.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: discrete TE in both directions with B=5 bins and lag 1. Output unit: nats. Plug-in bias under independence is about 0.0056 nats.

What you get: You get a directed information score in each direction.
What you do not get: You do not get a reliable ranking until bias or shuffle tests pass.

Formula

```
TE_{X->Y} = I(Y_{t+1}; X_past | Y_past)
```

Read it: Put your two return series into the formula. TE = Transfer entropy in nats; X_past, Y_past = Lagged binned returns; I(...|...) = Conditional mutual information. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
TE	Transfer entropy in nats
X_past, Y_past	Lagged binned returns
I(...|...)	Conditional mutual information

Pluses

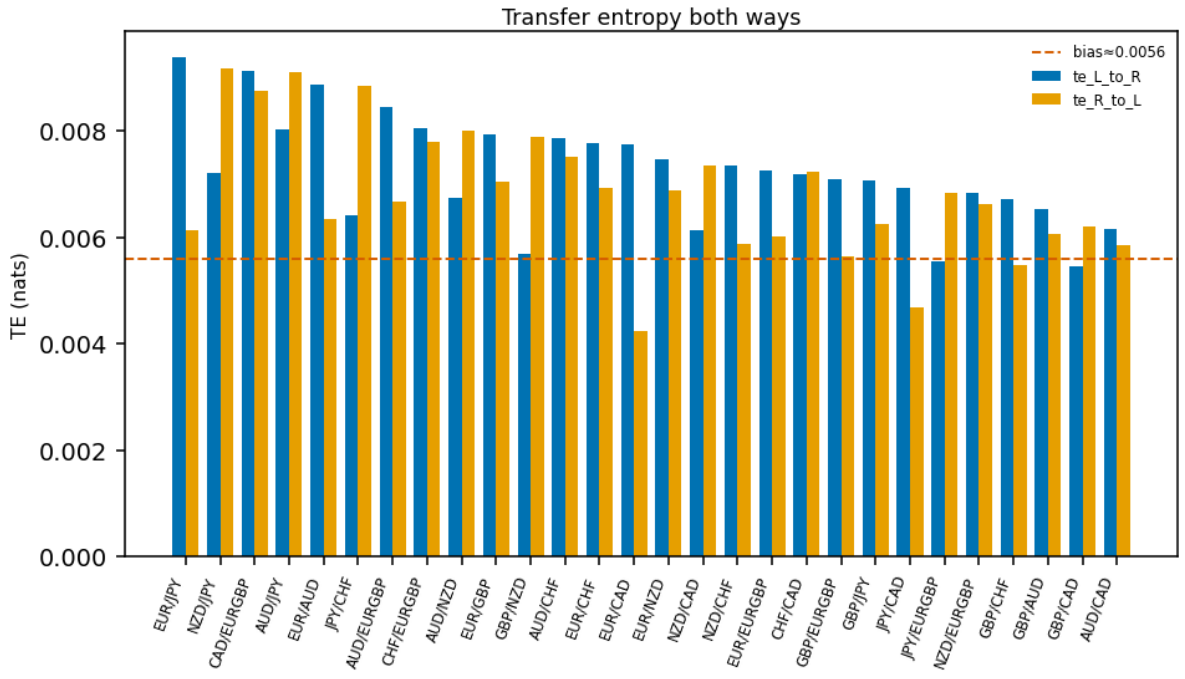
- Directed information flow in nats.
- Allows non-linear predictive content.
- Both directions are available.

Minuses

- Discrete TE has plug-in bias.
- Binning choices change the score.
- Many FX TE values sit near bias level.

Use it for: Directed screens after bias tests. Compare with Granger on the same lags.
Do not use it for: Claims without shuffle or surrogate tests. Exact crash timing.

Chart for this method



Directed scores. Peak |value|: EURUSD-USDJPY (te_L_to_R=0.0094, te_R_to_L=0.0061).

METHOD 18

Copulas

Question: What dependence structure links the uniform margins?

What this method is

A copula separates each margin from the dependence structure. Gaussian and Clayton families summarise different joint shapes.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: Gaussian copula rho and Clayton theta on empirical uniforms. Output unit: Gaussian rho in [-1, 1]; Clayton theta is a family parameter.

What you get: You get a dependence parameter after transforming returns to uniforms.
What you do not get: You do not get a full crash model unless you also study tails.

Formula

```
F_XY(x,y) = C( F_X(x), F_Y(y) )
C_Gauss(u,v; rho) = Phi_2( Phi^{-1}(u), Phi^{-1}(v); rho )
```

Read it: Put your two return series into the formula. u, v = Uniform transforms of the two returns; C = Copula function; rho = Gaussian copula dependence. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
u, v	Uniform transforms of the two returns
C	Copula function
rho	Gaussian copula dependence

Pluses

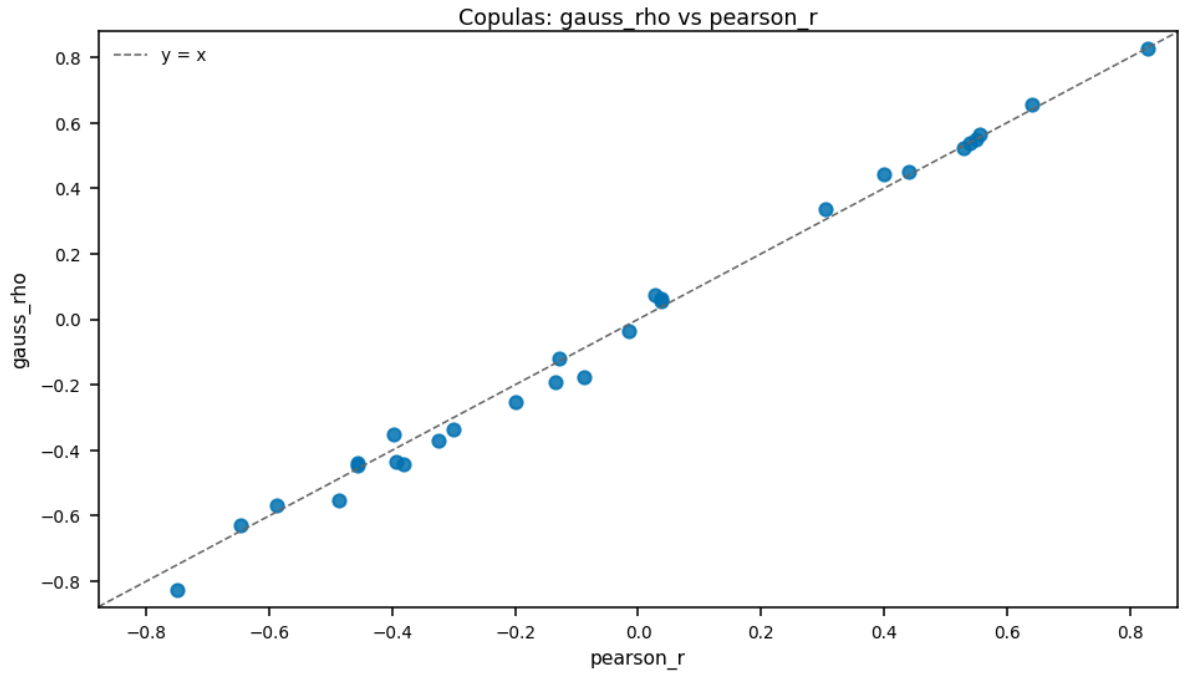
- Separates margins from dependence.
- Clayton and Gauss capture different tails.
- Kendall from copula aids comparison.

Minuses

- Model choice (Gauss vs Clayton) matters.
- Fitting needs careful uniforms u,v.
- Misspecified copulas misstate tails.

Use it for: Dependence after margin transforms. Bridge to tail dependence (method 19).
Do not use it for: Raw return Pearson replacement without care. Unsigned crash risk alone.

Chart for this method



Each point is one FX pair. Largest |gauss_rho – pearson_r|: AUDUSD-USDJPY (-0.177 vs -0.088).

METHOD 19

Tail Dependence

Question: How often do both legs sit in the extreme lower tail?

What this method is

Tail dependence counts how often both series are in an extreme quantile together. Lower and upper tails can differ.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: empirical L_hat at alpha=0.05 for lower and upper tails, with event counts. Output unit: conditional frequency in [0, 1].

What you get: You get a countable crash (or rally) co-occurrence rate.
What you do not get: You do not get the mathematical limit lambda unless alpha goes to zero.

Formula

```
L_hat(alpha) = (# u<=alpha AND v<=alpha)
/ (# v<=alpha)
```

Read it: Put your two return series into the formula. u, v = Empirical uniforms of the returns; alpha = Tail cutoff (0.05 here); L_hat = Conditional joint-tail frequency. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
u, v	Empirical uniforms of the returns
alpha	Tail cutoff (0.05 here)
L_hat	Conditional joint-tail frequency

Pluses

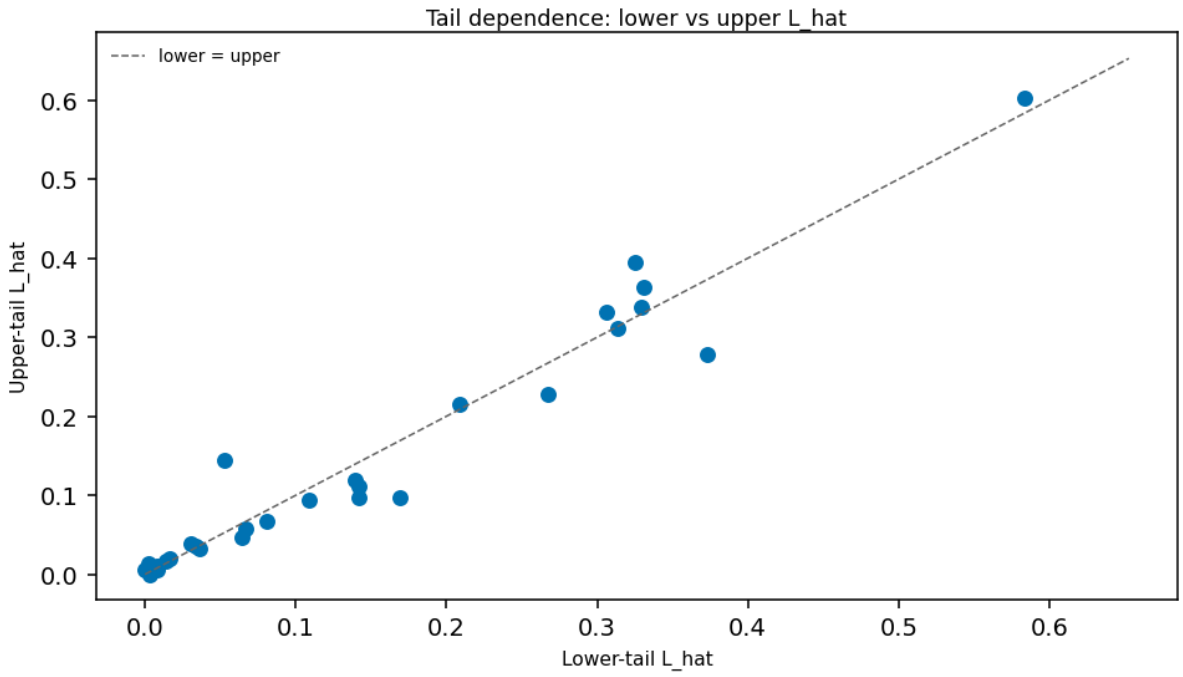
- Direct count of joint crash days.
- Lower and upper tails can differ.
- Easy to audit with num/den counts.

Minuses

- Finite alpha is not the limit lambda.
- Benchmark should be Gaussian, not independence.
- Rare events give noisy ratios.

Use it for: Crash co-movement vs average Pearson r. Carry pairs with hidden lower-tail risk.
Do not use it for: Daily trading signals alone. Claims without a same-r Gaussian benchmark.

Chart for this method



Lower vs upper tail at alpha=0.05. Strongest lower tail: AUDUSD-NZDUSD = 0.583.

METHOD 20

Cross-Quantilogram

Question: Do quantile hits of X predict quantile hits of Y at lag k?

What this method is

The cross-quantilogram correlates indicator hits of chosen quantiles across time. It focuses on extreme or moderate quantile dependence with lags.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: rho_tau(k) at tau in {{0.05, 0.20}} and lags 0..10; headline tau=0.05, k=1. Output unit: quantilogram correlation in [-1, 1].

What you get: You get directed extreme-quantile dependence at a chosen lag.
What you do not get: You do not get a center-of-distribution Pearson substitute.

Formula

```
h_x,t = 1(x_t < q_x) - tau
rho_tau(k) = SUM h_x,t h_y,t+k
/ sqrt( SUM h_x^2 * SUM h_y,t+k^2 )
```

Read it: Put your two return series into the formula. tau = Quantile level (0.05 in the headline); h_x,t = Centered quantile hit; rho_tau(k) = Cross-quantilogram at lag k. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
tau	Quantile level (0.05 in the headline)
h_x,t	Centered quantile hit
rho_tau(k)	Cross-quantilogram at lag k

Pluses

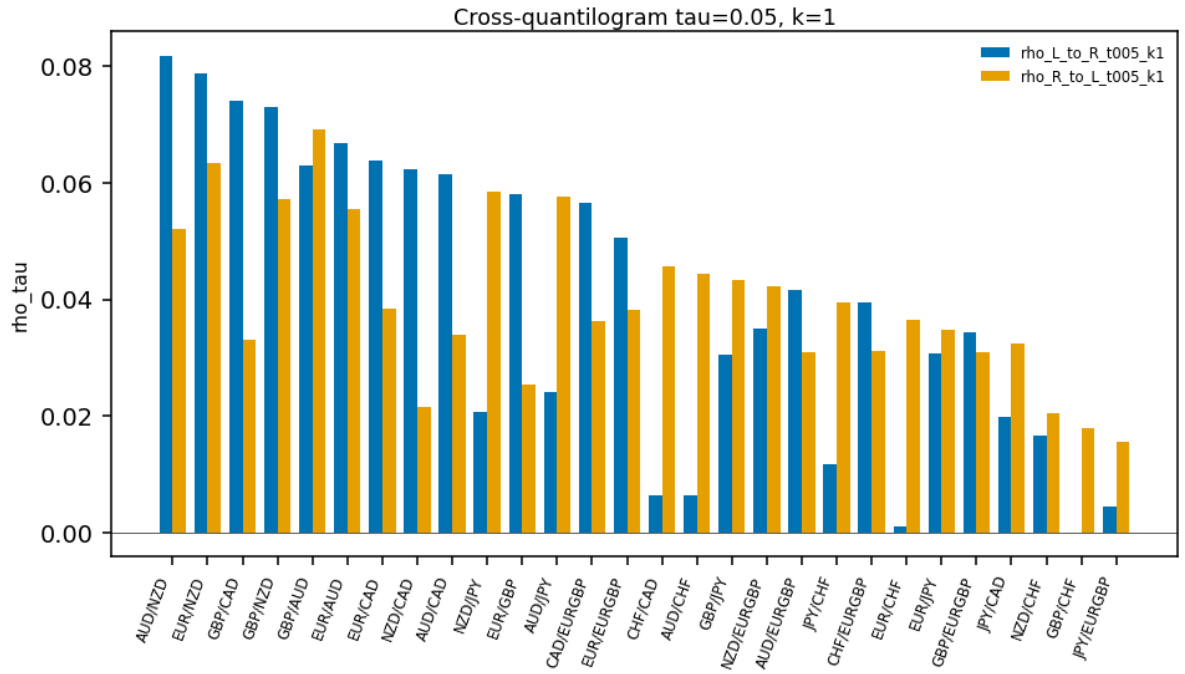
- Focuses on chosen quantile tau.
- Lag k gives directed extreme dependence.
- Useful for left-tail (tau=0.05) risk.

Minuses

- Needs expanding quantile estimates.
- Many tau and k raise false hits.
- Inference needs careful critical values.

Use it for: Tail-to-tail predictability screens. Complement to static L_hat tails.
Do not use it for: Center-of-distribution Pearson only. Untested multi-lag fishing.

Chart for this method



Directed scores. Peak |value|: AUDUSD-NZDUSD (rho_L_to_R_t005_k1=0.0818, rho_R_to_L_t005_k1=0.0520).

METHOD 21

Spectral Coherence

Question: At which cycle lengths do the series share power?

What this method is

Spectral coherence measures shared power at each frequency. It shows whether co-movement is short-cycle or long-cycle.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: magnitude-squared coherence at periods near 5, 21 and 63 days. Output unit: gamma^2 in [0, 1].

What you get: You get co-movement strength by cycle length.
What you do not get: You do not get a same-day crash count or a signed hedge ratio.

Formula

```
gamma^2(f) = |S_xy(f)|^2 / ( S_xx(f) * S_yy(f) )
```

Read it: Put your two return series into the formula. f = Frequency (cycles per day); S_xx, S_yy = Auto-spectra; S_xy = Cross-spectrum. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
f	Frequency (cycles per day)
S_xx, S_yy	Auto-spectra
S_xy	Cross-spectrum

Pluses

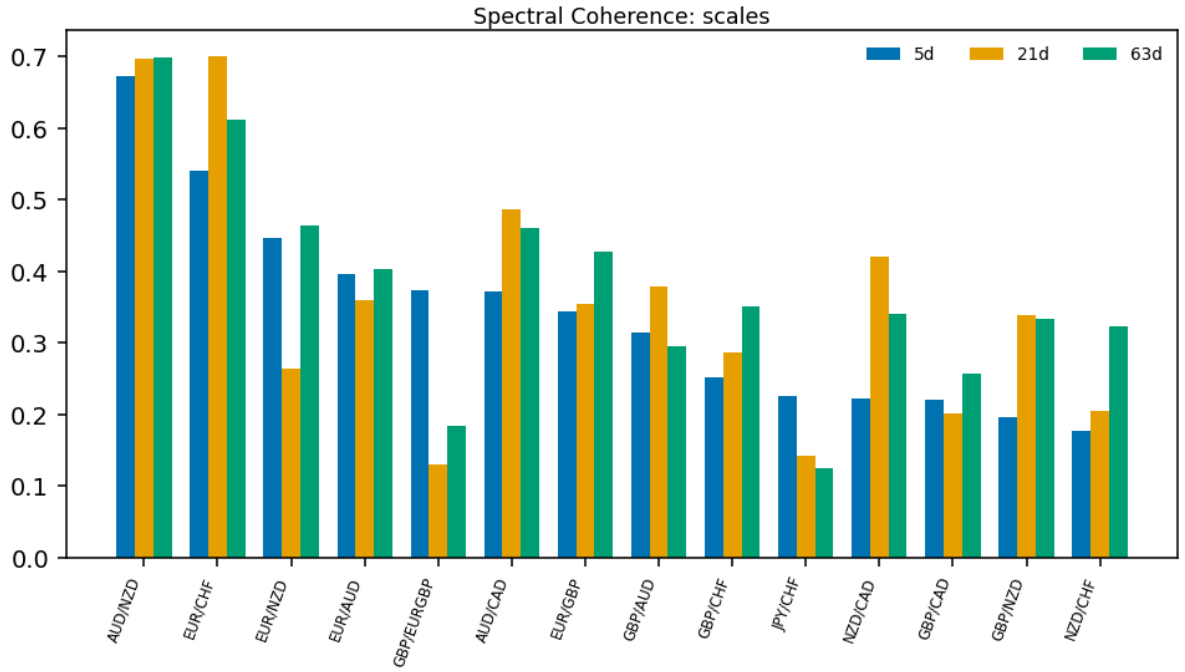
- Splits dependence by frequency.
- gamma^2 lies in [0, 1].
- Phase shows relative timing at f.

Minuses

- Stationarity assumptions matter.
- Window choice affects the spectrum.
- Hard to map one f to a trade rule.

Use it for: Cycle co-movement at 5 / 21 / 63 days. Checks whether short and long cycles differ.
Do not use it for: Same-day crash counts. Signed hedge ratios alone.

Chart for this method



Grouped scales. Example head pair AUDUSD-NZDUSD: 5d=0.673, 21d=0.696, 63d=0.699.

METHOD 22

Wavelet Correlation and Coherence

Question: How does correlation change across wavelet time scales?

What this method is

Wavelet correlation measures co-movement at chosen time scales. It localises dependence in both time and horizon.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: Morlet wavelet correlation at periods 5, 21 and 63 days. Output unit: correlation-like rho in [-1, 1].

What you get: You get multi-horizon co-movement for each pair.
What you do not get: You do not get a single desk KPI without picking a scale.

Formula

```
rho_wavelet(s) = Corr( W_x(s,·), W_y(s,·) )
```

Read it: Put your two return series into the formula. s = Wavelet scale (maps to day period); W_x, W_y = Wavelet coefficients at scale s; rho_wavelet = Correlation of those coefficients. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
s	Wavelet scale (maps to day period)
W_x, W_y	Wavelet coefficients at scale s
rho_wavelet	Correlation of those coefficients

Pluses

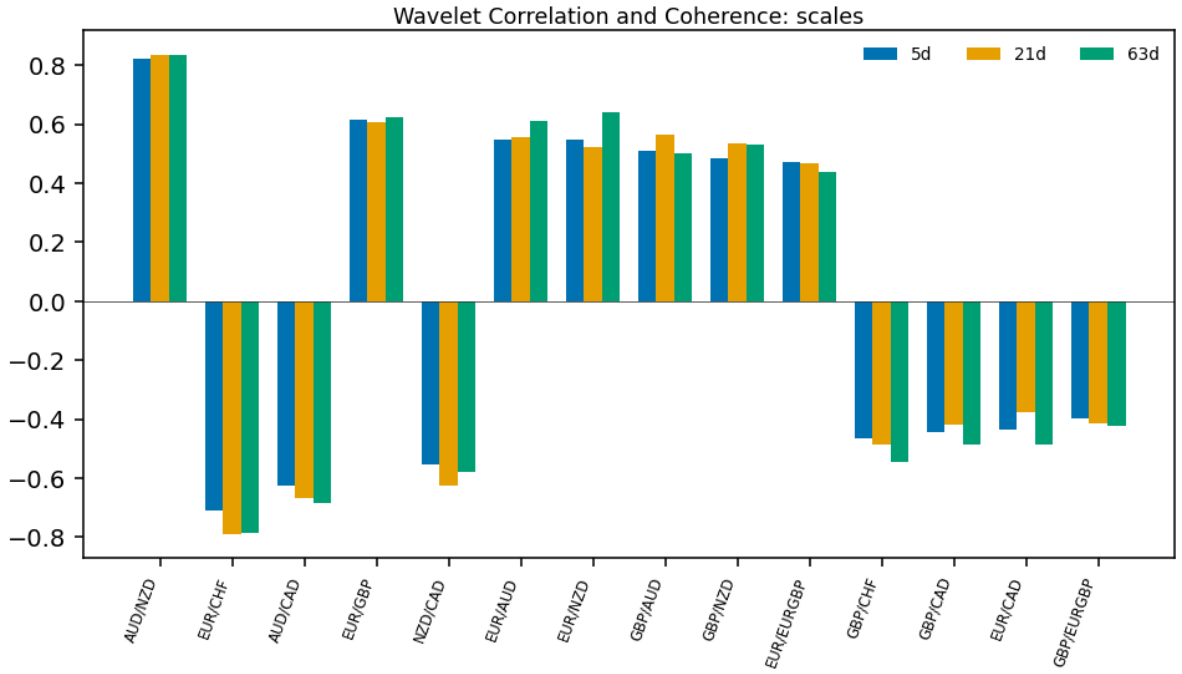
- Localises dependence in time and scale.
- Morlet scales map to day periods.
- Mean R^2 shows coherence strength.

Minuses

- Edge effects need a cone of influence.
- Scale choice is a design decision.
- Harder to audit than Pearson r.

Use it for: Multi-horizon co-movement maps. Compare rho at 5, 21 and 63 days.
Do not use it for: Single-number desk headlines alone. Ignoring edge masks.

Chart for this method



Grouped scales. Example head pair AUDUSD-NZDUSD: 5d=0.822, 21d=0.836, 63d=0.833.

METHOD 23

DCCA Detrended Cross-Correlation

Question: After local detrend, how do profiles co-move at scale s?

What this method is

DCCA builds cumulative return profiles, removes local trends, then correlates the residuals. Scale s shows short versus longer memory links.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: rho_DCCA at scales 20, 60 and 100. Output unit: DCCA correlation in [-1, 1].

What you get: You get detrended co-movement at chosen memory scales.
What you do not get: You do not get lead-lag direction from this score alone.

Formula

```
rho_DCCA(s) = mean(X~ Y~)
/ sqrt( mean(X~^2) * mean(Y~^2) )
```

Read it: Put your two return series into the formula. X~, Y~ = Detrended profile residuals in windows of length s; s = Window scale in observations; rho_DCCA = Detrended cross-correlation. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
X~, Y~	Detrended profile residuals in windows of length s
s	Window scale in observations
rho_DCCA	Detrended cross-correlation

Pluses

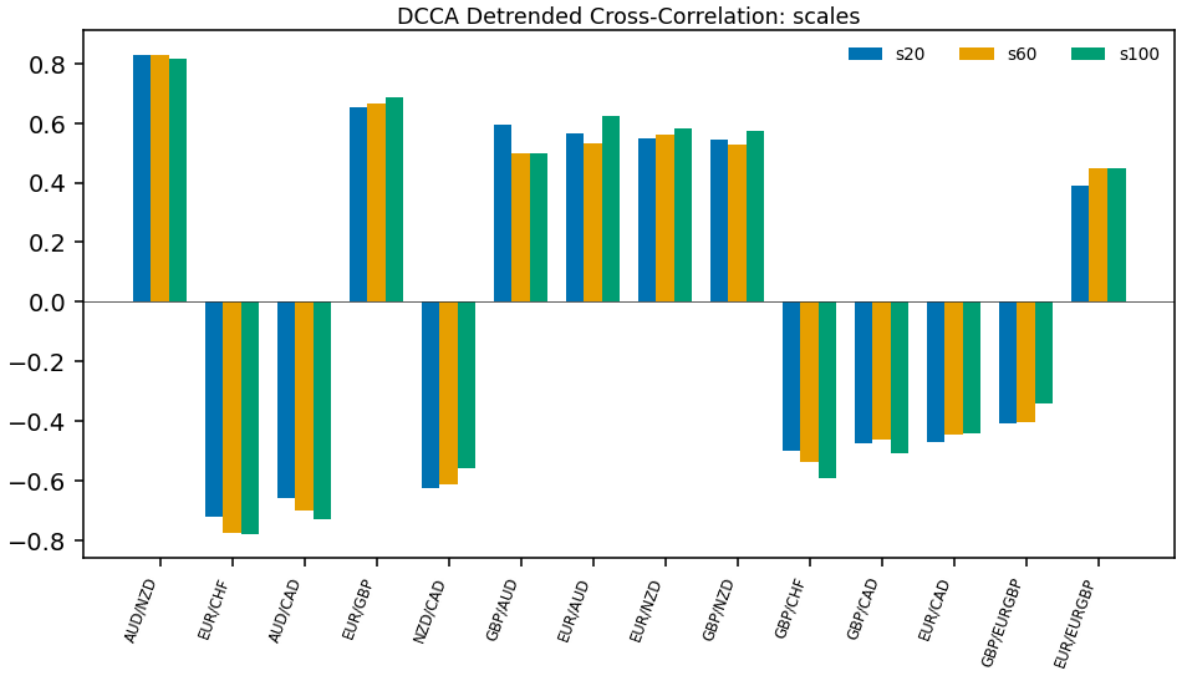
- Removes local trends before correlation.
- Scale s shows short vs long memory links.
- Useful when trends dominate raw r.

Minuses

- Window and poly order are free choices.
- Interpretation needs care vs Pearson.
- Cost rises with many scales.

Use it for: Long-memory co-movement screens. Compare rho_s20 vs rho_s100.
Do not use it for: Same-day crash frequency. Lead-lag direction alone.

Chart for this method



Grouped scales. Example head pair AUDUSD-NZDUSD: s20=0.828, s60=0.829, s100=0.815.

METHOD 24

Multifractal DCCA

Question: How does DCCA correlation change with moment order q?

What this method is

Multifractal DCCA weights fluctuation sizes with moment q. q=2 links to standard DCCA; higher q emphasises large moves.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: rho_q at selected q and scales; headline q=2, s=100. Output unit: multifractal DCCA correlation in [-1, 1].

What you get: You get scale-and-moment dependence for each pair.
What you do not get: You do not get a simple one-number desk KPI without fixed q and s.

Formula

```
G_xy(q,s) = mean[ sign(c) * |c|^(q/2) ]
rho_q(s) = G_xy / sqrt( G_xx * G_yy )
```

Read it: Put your two return series into the formula. q = Moment order; c = Local fluctuation product; rho_q(s) = Multifractal DCCA correlation. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
q	Moment order
c	Local fluctuation product
rho_q(s)	Multifractal DCCA correlation

Pluses

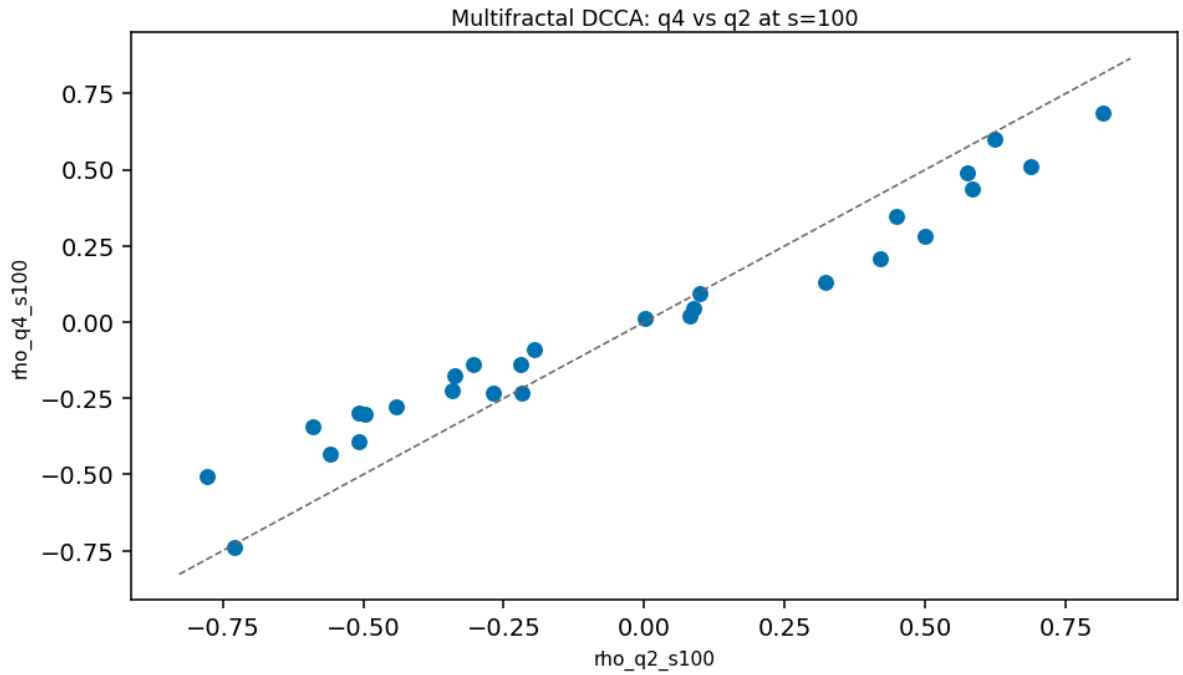
- q emphasises small or large fluctuations.
- q=2 links back to standard DCCA.
- Helps multifractal co-movement maps.

Minuses

- q and s add many free choices.
- Harder for non-specialist readers.
- Needs stable long samples.

Use it for: Scale and moment dependence together. Compare rho_q2 and rho_q4 at s=100.
Do not use it for: Simple one-page desk KPIs alone. Tiny samples.

Chart for this method



Moment q=4 vs q=2 at s=100. Largest |gap|: EURUSD-USDCHF (q4=-0.508, q2=-0.777).

METHOD 25

Volatility and Realized Correlation

Question: Do absolute returns (volatility proxies) move together?

What this method is

Volatility correlation is Pearson r of absolute returns. It tracks whether quiet and loud days line up across pairs.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: rho_vol = Corr(|x|, |y|) on daily abs returns. Output unit: correlation in [-1, 1].

What you get: You get a vol-vol linkage score on D1 closes.
What you do not get: You do not get true intraday realised correlation from daily closes alone.

Formula

```
rho_vol = Corr( |x_t|, |y_t| )
```

Read it: Put your two return series into the formula. |x_t|, |y_t| = Absolute daily log returns; rho_vol = Correlation of the volatility proxies. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
|x_t|, |y_t|	Absolute daily log returns
rho_vol	Correlation of the volatility proxies

Pluses

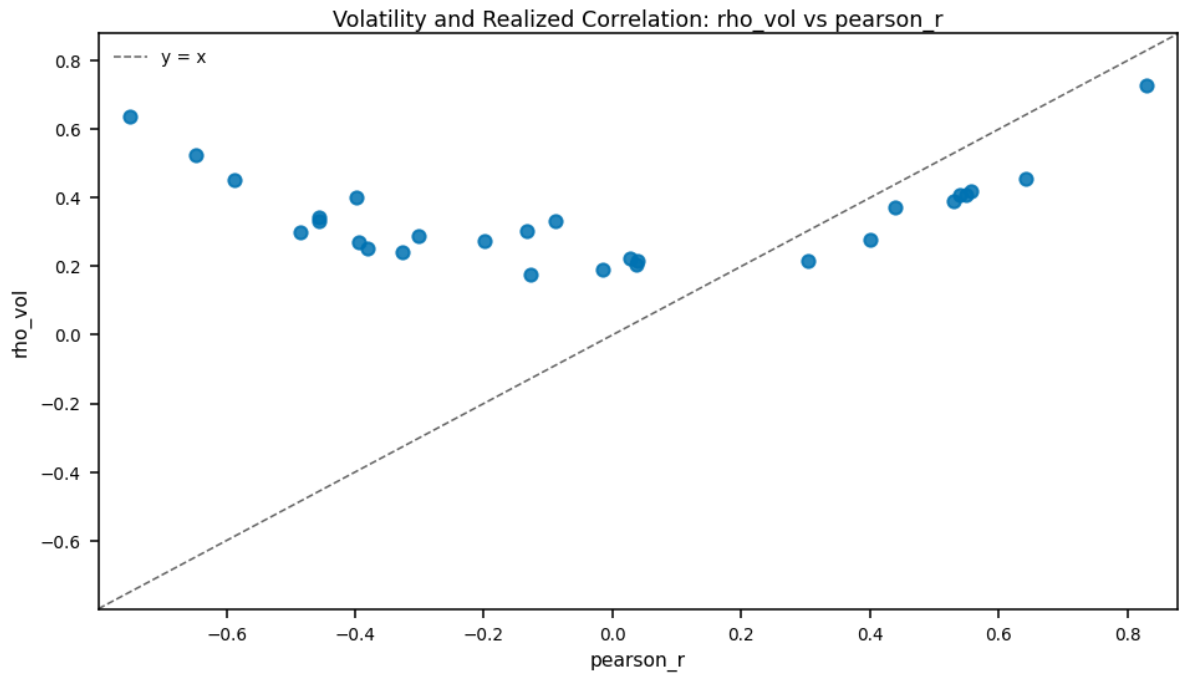
- Captures vol-vol co-movement.
- Simple on daily closes.
- Useful when |r| is low but vol clusters.

Minuses

- Absolute returns are not signed returns.
- True realized corr needs intraday buckets.
- D1 proxy is not full realized covariance.

Use it for: Volatility linkage screens on D1. Risk when vols rise together.
Do not use it for: Signed return hedging. Calling D1 |r| true intraday realized corr.

Chart for this method



Each point is one FX pair. Largest |rho_vol – pearson_r|: EURUSD-USDCHF (0.636 vs -0.749).

METHOD 26

Hayashi-Yoshida Covariance

Question: How do you covary returns when ticks are not synchronous?

What this method is

Hayashi-Yoshida covariance sums products of returns whose time intervals overlap. It targets asynchronous trading.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: HY covariance and a correlation form rho_hy on the aligned D1 panel. Output unit: correlation-like rho_hy in [-1, 1].

What you get: You get an asynchronous-aware covariance summary (limited power on pure D1).
What you do not get: You do not unlock full HY value without true tick intervals.

Formula

```
C_HY = SUM_i SUM_j dX_i * dY_j * 1(I_i overlaps J_j)
```

Read it: Put your two return series into the formula. dX_i, dY_j = Returns on intervals I_i and J_j; 1(overlap) = Indicator that intervals overlap; C_HY = Hayashi-Yoshida covariance. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
dX_i, dY_j	Returns on intervals I_i and J_j
1(overlap)	Indicator that intervals overlap
C_HY	Hayashi-Yoshida covariance

Pluses

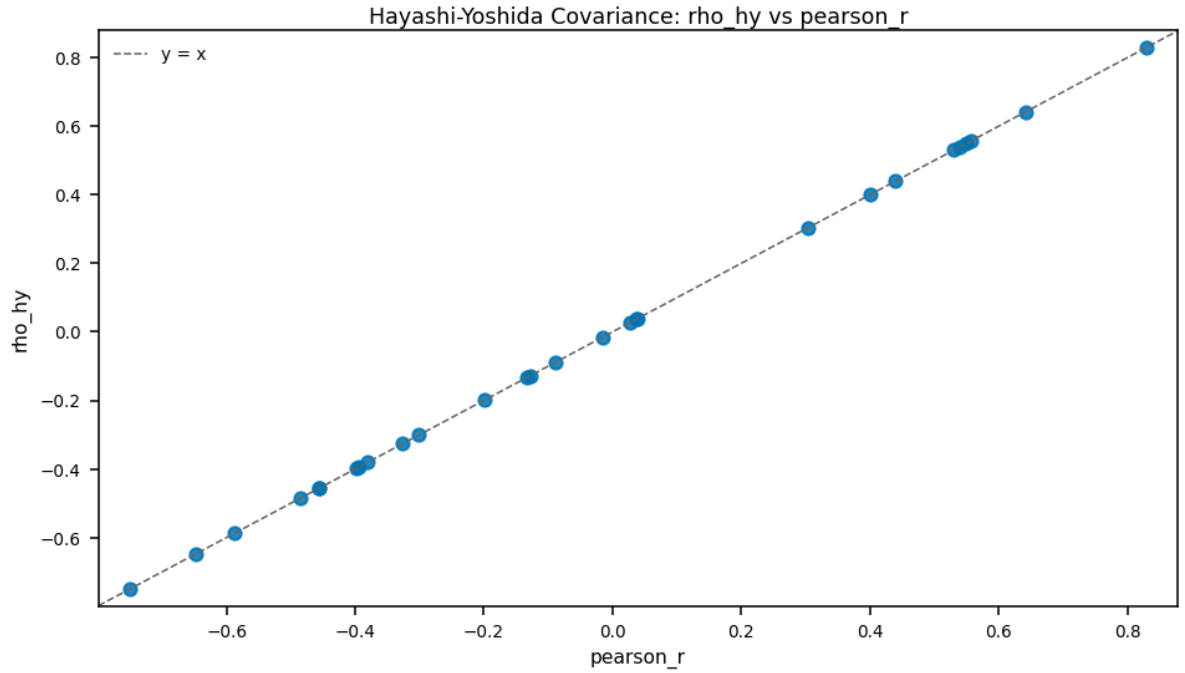
- Handles non-synchronous trading times.
- Classic for irregular tick data.
- Avoids forced grid refresh bias.

Minuses

- Needs true tick intervals for full power.
- On D1 closes the gain is limited.
- Harder to implement than Pearson.

Use it for: Intraday or irregular sampling designs. Compare rho_hy with daily Pearson.
Do not use it for: Only daily close panels as the main case. Lead-lag information flow alone.

Chart for this method



Each point is one FX pair. Largest |rho_hy – pearson_r|: USDJPY-USDCHF (0.399 vs 0.399).

METHOD 27

Cointegration and Error Correction

Question: Do log prices share a mean-reverting spread?

What this method is

Cointegration asks whether two log price levels stay linked by a stationary spread. An error-correction term kappa shows how fast the spread pulls back.

How we use it in this project

Input: log closes for Engle-Granger; log returns for the ECM side check. Snapshot 2026-11-03_majors_alpari_D1. Statistic: cointegration p-value and ECM kappa. Output unit: p-value; kappa is a speed coefficient.

What you get: You get a level-link test, which is different from return correlation.
What you do not get: You do not get a proven link from a raw p<0.05 across 28 pairs without multiplicity control.

Formula

```
e_t = ln(PY_t) - alpha - beta*ln(PX_t)
delta ln(PY_t) = c + kappa*e_{t-1} + lags + u_t
```

Read it: Put your two return series into the formula. e_t = Spread residual of log prices; kappa = Error-correction speed; coint_p = Engle-Granger p-value. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
e_t	Spread residual of log prices
kappa	Error-correction speed
coint_p	Engle-Granger p-value

Pluses

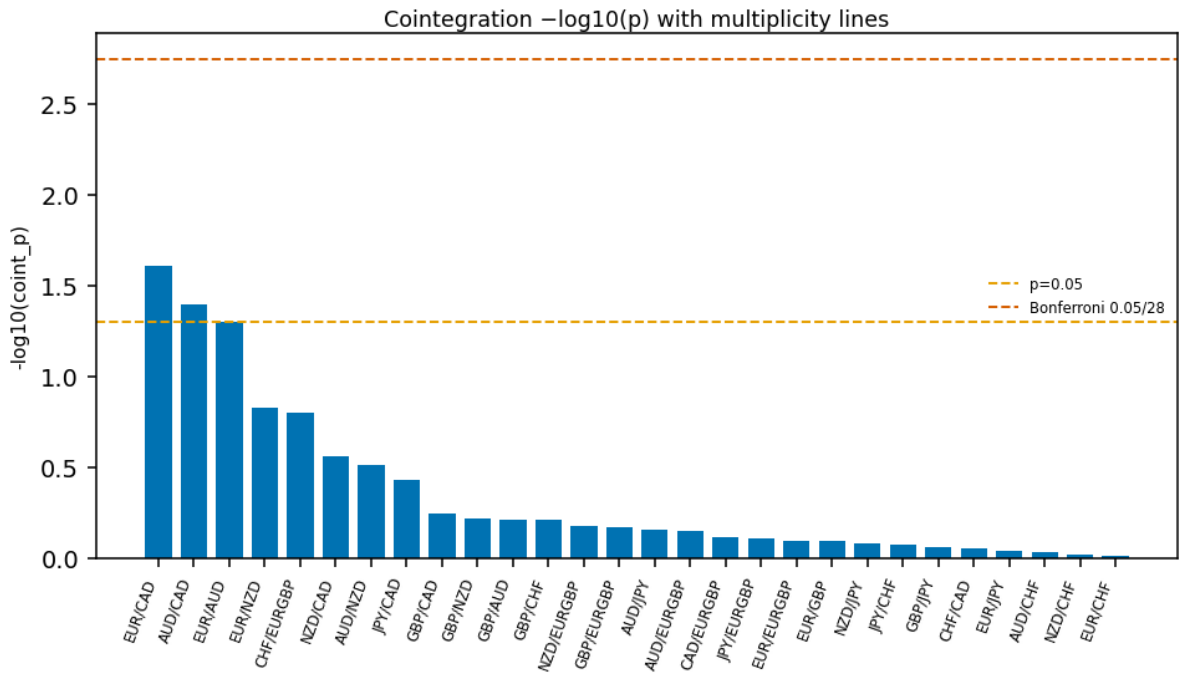
- Tests level links, not only return corr.
- kappa shows error-correction speed.
- Key for pairs and mean-reversion ideas.

Minuses

- Many pairs raise false positives.
- Bonferroni can wipe raw 5% hits.
- High return corr does not imply cointegration.

Use it for: Price-spread stationarity screens. Separate return corr from level tethering.
Do not use it for: Treating raw p<0.05 across 28 pairs as proven. Crash timing.

Chart for this method



Engle-Granger −log10(p). Lowest raw p: EURUSD-USDCAD = 0.0245. Bonferroni line is at p≈0.0018; no pair clears it in v001.

METHOD 28

Networks PCA and Connectedness

Question: How close are pairs in a network, and who transmits shocks?

What this method is

This method builds network distances from correlations, PCA of the correlation matrix, and Diebold-Yilmaz connectedness from a VAR forecast-error variance decomposition.

How we use it in this project

Input: daily log return of close on matched dates from snapshot 2026-11-03_majors_alpari_D1 (eight majors). Statistic: pair distance d_ij, directional theta, system TCI, and PCA eigenvalues. Output unit: d_ij >= 0; TCI in percent; variance shares in [0, 1].

What you get: You get a system view: closeness, factor share, and shock transmission.
What you do not get: You do not get a pair-trade entry rule from TCI alone.

Formula

```
d_ij = sqrt( 2*(1 - rho_ij) )
TCI = 100 * SUM_{i!=j} theta_tilde_ij / N
```

Read it: Put your two return series into the formula. rho_ij = Pearson correlation of the pair; d_ij = Network distance (smaller = closer); TCI = System total connectedness index. The result is the number this method stores for the pair.

What the symbols mean

Symbol	Meaning on this data
rho_ij	Pearson correlation of the pair
d_ij	Network distance (smaller = closer)
TCI	System total connectedness index

Pluses

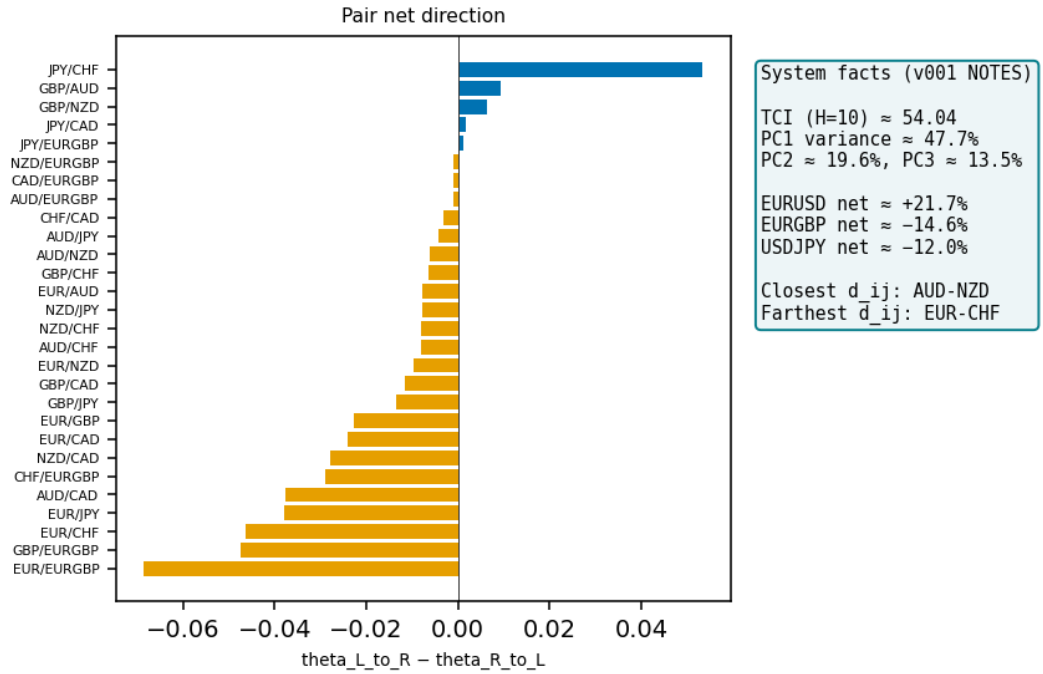
- System view beyond one pair.
- PCA shows shared factors.
- Diebold-Yilmaz TCI shows spillover intensity.

Minuses

- VAR connectedness needs lag choice.
- Near-singular panels (EURGBP) distort PCA.
- Full-sample TCI is not a forecast.

Use it for: Market-wide connectedness maps. Net transmitter vs receiver ranks.
Do not use it for: Pair trade entries without more filters. Ignoring triangular redundancy.

Chart for this method



Left: pair net spillover direction. Largest |net|: EURUSD-EURGBP = -0.068. Right: system PCA/TCI from v001 NOTES.